Release Determinacy for Hyperedge-Level Fairness Certification in Attributed Hypergraph Clustering (Extended Version)

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***Abstract*—Attributed hypergraph clustering (AHC) has been widely used to group entities linked by hyperedges, such as coauthor teams. Since decisions are often made per cluster, auditors check partitions for unequal treatment within hyperedges, often from a data release such as a pairwise graph rather than from the base hyperedge table. However, the same pairwise graph can arise from different sets of hyperedges, which a partition may split by demographic group to different degrees. As a result, an audit of this graph can pass a partition that exceeds a declared fairness budget. Judging every budget correctly from a release alone requires the release to determine the audit query, which we call release determinacy. Though fairness auditing has been studied for clustering and view determinacy for relational queries, release determinacy of fairness audits is rarely considered. Therefore, we study the novel problem of release determinacy for fairness certification in AHC. The key challenge is to find such releases and to bound the audit value for the others. To address this challenge, we first define hyperedge group preservation (HGP), an additive aggregate that measures the separation of groups across clusters within each hyperedge. Our theoretical analysis shows that the random-walk release, a weighted pairwise graph computed by AHC methods, determines HGP for hyperedges of up to three members. For larger hyperedges, where pairwise graphs cannot determine HGP in general, the random-walk release still yields an upper bound in one pass, called a certificate, and computing the tightest certificate is NP-hard. For the budgets left open, hyperedge-level counts computed by one SQL query fix the verdict exactly. Extensive experiments on six real-world datasets show that audits of the pairwise graph misjudge partitions in both directions, whereas the random-walk certificate passes most partitions within the budget and none above it.**

***Index Terms*—Hypergraph Clustering, Fairness Certification, View Determinacy, Sufficient Statistics, Responsible Data Management**

Artifact Availability: The source code, data, and/or other artifacts have been made available at https://github.com/yingkongshibetter/release-determinacy-ahc-artifact.

I. Introduction

Many datasets record interactions among more than two entities, such as coauthored papers, households, and film casts, which hypergraphs represent as hyperedges. Attributed hypergraph clustering (AHC) groups the nodes of such hypergraphs by their hyperedges and attributes [1]–[5]. Decisions are then often made per cluster, such as inviting one cluster of researchers to a program or sending the same recommendations to one cluster of users. If a partition separates the members of one team by demographic group, a decision per cluster treats these members unequally, for example by inviting only some of them. Detecting such disparities requires independent audits of the partition.

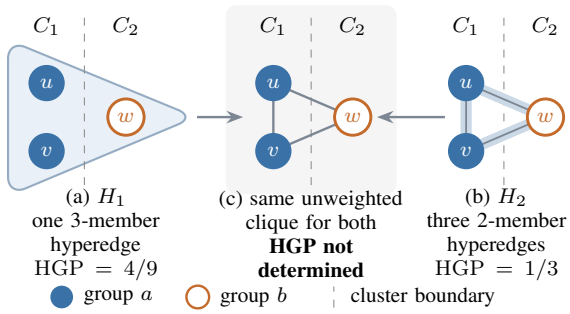


Fig. 1. Running example. (a) One three-author paper. (b) Three two-author papers. (c) Their common unweighted clique.

Audits of algorithmic systems often rely on releases, the data that the audited organization provides, rather than on its raw records [6], [7]. Since common graph analysis, learning, and debiasing libraries operate on pairwise edges [8]–[10], the release for AHC can be a pairwise graph of the node pairs sharing a hyperedge. Such a graph keeps the pairs but drops the hyperedges behind them. For example, a three-author paper and three two-author papers by the same researchers yield the same pairwise graph, as shown in Figure 1. A partition that separates these researchers by group splits the whole team in the first case but only two of the three pairs in the second. An audit of the pairwise graph cannot tell the two cases apart and can pass a partition above a budget, an upper limit that a policymaker declares on the audit value.

This example raises a data-management question of release design. In database terms, the release is a view of the base hyperedge table, the audit is an aggregate query over this table joined with the attributes and the partition, and release determinacy is the determinacy of this query by the view [11], [12]. The data owner runs the clustering and holds the hyperedges. The auditor holds the protected attributes, which organizations often store separately from operational records [13], [14]. With a release alone, neither party receives the data of the other, although an exact audit value needs both. Releases of multiway data range from the unweighted clique expansion, the unweighted clique for short [15]–[19],

to releases with pair weights, such as weighted coauthorship networks [20], [21] and the random-walk release that AHC methods compute [3], [4]. The random-walk release gives the hypergraph transition probabilities between nodes together with the node degrees. A release supports certifying a budget in two cases. It either fixes the audit value for every hypergraph that produces the release or yields an upper bound valid for all such hypergraphs. We call the first property release determinacy.

However, certifying the fairness of AHC from a release is a non-trivial task. First, the audit must capture disparities within hyperedges, such as those inside a household or a team [22], [23]. Existing fair clustering methods and auditing tools [24]–[27] instead measure balance across clusters and assume an observable audit value. Second, the releases that determine such an audit are unknown. Recent AHC methods [3], [4] compute random-walk releases for clustering quality rather than auditability. View determinacy theory [11], [12], [28] mostly studies conjunctive queries rather than fairness aggregates. Third, a release that does not determine the audit value can still bound it from above. Yet a loose bound leaves budgets undecided. Deciding these budgets exactly requires hyperedge-level statistics.

Therefore, in this paper, we study the novel problem of release determinacy for fairness certification in AHC. To address these challenges, we first define hyperedge group preservation (HGP), a segregation measure that compares the spread of each group of a hyperedge over clusters with that of the whole hyperedge. A budget on HGP limits the average gap that cluster-level decisions create within hyperedges. We then bound the HGP of each hyperedge by its crossing pairs, the pairs of members that differ in both group and cluster. Up to three members, the crossing pairs capture the disparity of a hyperedge exactly. The random-walk release thus determines HGP at these sizes, even when sizes are mixed. From four members on, no release built from pair counts determines HGP in general. For these larger hyperedges, the bound yields conservative certificates that never understate HGP and need only the release. For the budgets left undecided, the auditor requests hyperedge-level counts that fix the verdict exactly. We call this combination certify-then-request.

Our main contributions are as follows.

- We formulate the problem of release determinacy for fairness certification in AHC and introduce HGP.
- We locate a sharp determinacy boundary of HGP between hyperedge sizes three and four for weighted clique expansions and the random-walk release. Up to three members, the random-walk release determines HGP even on multihypergraphs with mixed sizes, where no weighted clique expansion does.
- We derive conservative certificates from the random-walk release and the normalized clique expansion [29] as dual solutions of a linear relaxation of the largest HGP that the release allows. The random-walk certificate needs no cap on hyperedge size and, with the node degrees, remains conservative on rounded or pruned releases. Computing the tightest certificate is NP-hard, and the hyperedge-level counts that decide the remaining budgets are computed by one SQL query that scales to hundreds of millions of memberships.
- Experiments on real-world datasets show that auditors of pairwise graphs misjudge partitions in both directions where hyperedges overlap. The unweighted clique leaves many verdicts undetermined even when the number and sizes of hyperedges are declared. The random-walk certificate certifies most passing partitions without access to the hyperedges.

II. Problem Setting

In this section, we define releases, audits, and the problem.

A. Attributed Hypergraphs and Releases

An *attributed hypergraph* $\mathcal{H} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ has n nodes, node attributes $\mathbf{X}$, and $m > 0$ hyperedges $e \subseteq \mathcal{V}$ with $|e| \geq 2$. A hypergraph is *simple* if all hyperedges are distinct, a *multihypergraph* if duplicates are allowed, and *r-uniform* if $|e| = r$ for all e. The volume is $\mathrm{vol}(\mathcal{H}) = \sum_e |e|$. An audited attribute A has groups $g \in G_A$, and a clustering $\mathcal{C}$ is a partition of $\mathcal{V}$. In the running example, nodes are authors, hyperedges are teams, groups $\{a, b\}$, and clusters C_1, C_2.

A *release* is the set of tables an auditor receives, in addition to the attributes and the partition. We use five categories of release, from no hyperedge structure to all of it.

- A *node-preserving release* retains node identifiers, attributes, and the partition but no hyperedge structure.
- An *attribute-similarity graph* $S = \mathrm{sim}(\mathbf{X})$ uses features alone, such as the cosine-kNN graphs of AHCKA [3] and AHRC [4].
- A *pairwise graph* is either an unweighted clique, which lists every pair contained in at least one hyperedge and drops hyperedge identifiers [29], or a weighted clique expansion, the *weighted clique* for short, which gives each pair the weight $w_{uv} = \sum_{e \supseteq \{u,v\}} \omega(|e|)$ for a declared rule ω. The *normalized clique expansion* [29], the *normalized clique* for short, is the weighted clique expansion with $\omega(k) = 1/(k-1)$ and no self-loops.
- A *random-walk release* gives $P = \mathbf{M}\,\mathrm{diag}(|e|)^{-1}\mathbf{M}^\top$, where $\mathbf{M}$ is the node–hyperedge membership matrix, so that $P_{uv} = \sum_{e \supseteq \{u,v\}} 1/|e|$ including the diagonal. It has at most $\sum_e |e|^2$ nonzero entries and takes $O(\sum_e |e|^2)$ time to build. Equivalently, it gives the hypergraph transition matrix $T = \mathrm{diag}(d)^{-1}P$ together with the node degrees d_v. AHRC and AHCKA compute T, and AHRC can prune its entries.
- A *hyperedge-preserving release* is either the incidence release, which lists node–hyperedge memberships with hyperedge identifiers, or the contingency table Ψ_A, which lists for each hyperedge e, group g of A, and cluster C the count $n_{e,g,C} = |\{v \in e \cap C : A(v) = g\}|$.

Pairwise graphs and the random-walk release lie between these extremes, as they keep counts of the pairs inside hyperedges but not the hyperedges themselves. We list the source, determinacy condition, and certificate of each release in Table I.

TABLE I
COMMON RELEASES, EXAMPLE SOURCES, THE INPUTS ON WHICH EACH RELEASE DETERMINES HGP, AND ITS CERTIFICATES.

Release	Source	Determined for	Certificate
Unweighted clique	SNAP [15], [16]	disjoint hyperedges, §III-B	–
	hollywood-2011 [17]		
Weighted clique	ogbl-collab, $\omega \equiv 1$ [18]	uniform size 2 or 3, Cor. 1(i)	–
Normalized clique	cond-mat, rounded [20], [21]	uniform size 2 or 3, Cor. 1(i)	U_Q, $U_{Q,\bar{r}}$
Random-walk release	AHCKA [3]	size ≤ 3, Cor. 2(i)	U_P
	AHRC, pruned [4]	–	U_P
Hyperedge-preserving	on request	all inputs, §III-A	not needed

B. Two Audits

a) Cluster-level audit.: The first audit checks the balance of the clusters. Let p_g be the share of group g among all nodes and $p_{g|C}$ its share within cluster C. *Cluster group parity* (CGP) is

$$\mathrm{CGP}(\mathcal{C}; A) = \sum_{C \in \mathcal{C}} \frac{|C|}{n} \, \mathrm{TV}(p_{\cdot|C}, p), \tag{1}$$

where $\mathrm{TV}(p, q) = \frac{1}{2} \sum_i |p_i - q_i|$. CGP follows the statistical-parity notions of fair clustering [24]–[26].

b) Hyperedge-level audit.: Disparities within hyperedges are well documented, as in the unequal sharing of resources within households [22], [30] and the unequal credit given for joint work [23]. An audit of such disparities should be neutral, bounded, membership-weighted, and local, and we define hyperedge group preservation.

For a hyperedge e let $q_e(C) = |e \cap C|/|e|$ and $q_{e,g}(C) = n_{e,g,C} / \sum_{C'} n_{e,g,C'}$ be the cluster distributions of the whole hyperedge and of its members in group g, and $p_{g|e}$ the share of group g in e. *Hyperedge group preservation* (HGP) is

$$\begin{aligned} \mathrm{HGP}(\mathcal{C}; A) &= \sum_{e \in \mathcal{E}} \frac{|e|}{\mathrm{vol}(\mathcal{H})} \, \psi_e, \\ \psi_e &= \sum_{g:\, p_{g|e} > 0} p_{g|e} \, \mathrm{TV}(q_{e,g}, q_e). \end{aligned} \tag{2}$$

HGP measures the difference between the spread of each group of a hyperedge over clusters and that of the whole hyperedge, and larger values mean stronger separation. It is neutral, as each term ψ_e is zero exactly when every present group is spread like its hyperedge, including when the hyperedge remains in one cluster. It is bounded, as $q_e \geq p_{g|e} \, q_{e,g}$ gives $\mathrm{TV}(q_{e,g}, q_e) \leq 1 - p_{g|e}$, so $\psi_e \leq 1 - \sum_g p_{g|e}^2 < 1$ and budgets share one scale. HGP is membership-weighted and local, as it sums per-hyperedge terms with weights $|e|/\mathrm{vol}(\mathcal{H})$, so each membership counts once and a violation traces to hyperedges, groups, and clusters. HGP complements CGP rather than replacing it, as CGP still detects clusters whose group shares depart from the population when every hyperedge stays in one cluster.

Example 1 (Running example)**.** In H_1 the hyperedge $e = \{u, v, w\}$ has $p_{a|e} = 2/3$ and $p_{b|e} = 1/3$, with $\mathrm{TV}(q_{e,a}, q_e) = 1/3$ and $\mathrm{TV}(q_{e,b}, q_e) = 2/3$, so $\mathrm{HGP}(H_1) = (2/3)(1/3) + (1/3)(2/3) = 4/9$. In H_2 the pair $\{u, v\}$ has $\psi = 0$ and each of $\{u, w\}$ and $\{v, w\}$ has $\psi = 1/2$ with weight $2/6$, so $\mathrm{HGP}(H_2) = 2 \cdot (2/6)(1/2) = 1/3$.

HGP also admits a statistical reading. Let $p_e(g, C) = n_{e,g,C}/|e|$ be the share of the members of e that belong to group g and lie in cluster C, and write $p_e(g) = p_{g|e}$ and $p_e(C) = q_e(C)$. Then

$$\psi_e = \frac{1}{2} \sum_{g,C} \big| p_e(g, C) - p_e(g) \, p_e(C) \big|, \tag{3}$$

which holds as $p_{g|e} \, |q_{e,g}(C) - q_e(C)| = |p_e(g, C) - p_e(g) \, p_e(C)|$ for every group present in e and every C, so ψ_e is the total-variation distance between the joint group–cluster distribution of e and the product of its marginals. With the clusters as the areal units of segregation measurement and the members of e as the population, $\psi_e = I_e \, D_e$, where $I_e = 1 - \sum_g p_{g|e}^2$ is Simpson's interaction index and $D_e \in [0, 1]$ is the generalized dissimilarity index of Sakoda [31], [32]. HGP is thus the membership-weighted within-hyperedge segregation of groups across clusters, and with two groups ψ_e is the index of Karmel and MacLachlan [33]. Conditioning on hyperedges as strata with clusters as outcomes gives the conditional demographic disparity of Wachter et al. [34] for EU non-discrimination law.

c) Dissimilarity scale.: With the membership-weighted interaction $\bar{I} = \sum_e (|e|/\mathrm{vol}(\mathcal{H})) \, I_e$, the ratio $\bar{D} = \mathrm{HGP}/\bar{I}$ averages the indexes D_e with the weights $|e| \, I_e$, so it lies in $[0, 1]$ like a dissimilarity index, and we call it the membership-weighted dissimilarity. If $\bar{I} = 0$, each hyperedge has one group, so $\mathrm{HGP} = 0$, and we set $\bar{D} = 0$.

Equal CGP does not imply equal HGP, as shown below.

Remark 1 (Same-group exchanges)**.** Exchanging the clusters of two nodes of the same group preserves every cluster size $|C|$ and every count $n_{g,C}$ of members of group g in cluster C. Any audit that is a function of these counts, such as CGP, balance [24], or the statistical-parity gap of each cluster, retains its value. HGP instead depends on the counts $n_{e,g,C}$ within hyperedges, which the aggregate counts do not fix, as in Simpson's paradox [35]. Let a_1, a_2 belong to group a and b_1, b_2 to group b. The hyperedges $\{a_1, b_1\}$ and $\{a_2, b_2\}$ with the clusters $\{a_1, b_2\}$ and $\{a_2, b_1\}$ give CGP 0 and HGP $1/2$. Exchanging a_1 and a_2 leaves CGP at 0 and lowers HGP to 0.

Remark 2 (Refinement)**.** If a partition $\mathcal{C}'$ refines $\mathcal{C}$, then $\mathrm{HGP}(\mathcal{C}; A) \leq \mathrm{HGP}(\mathcal{C}'; A)$ and $\bar{D}(\mathcal{C}; A) \leq \bar{D}(\mathcal{C}'; A)$. Merging clusters maps the joint group–cluster distribution of each hyperedge and the product of its marginals to those of $\mathcal{C}$ through one map, which cannot increase their total-variation distance ψ_e, and $\bar{I}$ does not depend on the partition.

d) Dyadic surrogate.: The value of Eq. (2) computed from incidence is *true HGP*, written $\mathrm{HGP}_{\mathrm{true}}$, and its comparison with a budget B on HGP gives the *true verdict*. A graph tool

that sees only the pairwise graph can instead evaluate Eq. (2) on one size-2 hyperedge per pair of the unweighted clique, which gives the *dyadic surrogate*, written $\mathrm{HGP}_{\mathrm{sur}}$. The dyadic surrogate treats the triangle as the three two-author papers of H_2 and gives $\mathrm{HGP}_{\mathrm{sur}} = 1/3$ even when the hypergraph is H_1 with $\mathrm{HGP}_{\mathrm{true}} = 4/9$. We use it as a diagnostic baseline for audits that read only the pairwise graph.

C. Interpretation of HGP

a) Decision model.: HGP targets decisions that are made per cluster, such as invitations to a program, mentoring cohorts, or one recommendation slate per cluster. We model such a decision by a cluster-level invitation policy $\pi : \mathcal{C} \to [0,1]$, where $\pi(C)$ is the invitation probability of everyone in cluster C. For a hyperedge e, $\Phi_g(e;\pi) = \sum_C \pi(C) q_{e,g}(C)$ is the average invitation probability of group g within e, and $\Phi(e;\pi) = \sum_C \pi(C) q_e(C)$ that of e.

Proposition 1 (HGP bounds decision gaps)**.** *(a) For every hyperedge e and every group g present in e, $\max_{\pi\in[0,1]^{\mathcal{C}}} |\Phi_g(e;\pi) - \Phi(e;\pi)| = \mathrm{TV}(q_{e,g}, q_e)$. (b) For every cluster-level invitation policy $\pi \in [0,1]^{\mathcal{C}}$, the membership-weighted average gap*

$$\sum_e \frac{|e|}{\mathrm{vol}(\mathcal{H})} \sum_g p_{g|e} \, |\Phi_g(e;\pi) - \Phi(e;\pi)| \;\le\; \mathrm{HGP}.$$

Proof. (a) Write $\Phi_g(e;\pi) - \Phi(e;\pi) = \sum_C \pi(C)\,\Delta(C)$ with $\Delta(C) = q_{e,g}(C) - q_e(C)$. Both $q_{e,g}$ and q_e are distributions over the clusters, so the differences $\Delta(C)$ sum to zero, and their positive part and their negative part each have mass $\frac{1}{2}\sum_C |\Delta(C)| = \mathrm{TV}(q_{e,g}, q_e)$. The expression is linear in π, so its absolute value over $\pi \in [0,1]^{\mathcal{C}}$ is largest when $\pi(C) = 1$ on the positive coordinates or on the negative ones, and this gives $\mathrm{TV}(q_{e,g}, q_e)$.

(b) Fix π and apply (a) to every hyperedge and group. Multiplying the bound for (e,g) by $p_{g|e}\,|e|/\mathrm{vol}(\mathcal{H})$ and summing gives the weighted average gap on the left and HGP on the right by Eq. (2). □

In particular, $\mathrm{HGP} \le B$ guarantees, for every such policy, that the membership-weighted average gap between the invitation probability of a group within a hyperedge and that of the whole hyperedge is at most B. With two clusters, the largest membership-weighted average gap of a single policy equals HGP, as one policy then attains all maxima of part (a). With more clusters, it is at most HGP.

A budget on $\bar{D}$ has a reading in terms of gaps between two groups. Let M_e be the largest total-variation distance between the cluster distributions of two groups present in e, which by part (a) is the largest gap that a policy can create between two groups of e. As q_e mixes the $q_{e,h}$ with the weights $p_{h|e}$, convexity gives $\mathrm{TV}(q_{e,g}, q_e) \le (1 - p_{g|e}) M_e$, so $\psi_e \le I_e M_e$ and $D_e \le M_e$. Hence $\bar{D} > \tau$ implies an average M_e above τ under weights $|e|\,I_e$.

b) Protection target.: An application that protects every group of every hyperedge audits the worst gap $\max_{e,g} \mathrm{TV}(q_{e,g}, q_e)$. By Proposition 1(a), it is the largest gap a cluster-level policy can create for any group in any hyperedge, which the contingency table fixes.

D. Problem Statement

Certification needs inputs with equal releases to share the audit value.

Definition 1 (Determinacy)**.** Fix a class of admissible attributed hypergraphs. A release *determines* an audit on this class if inputs with equal release, attributes, and partition have equal audit values.

The class can contain all attributed hypergraphs or only those with a declared hyperedge size, and such metadata belongs to the class, not to the release. If a release does not determine HGP, some budget passes one input that produces it and fails another. For the running example, every budget in $[1/3, 4/9)$ passes H_2 and fails H_1.

Formally, the data owner holds the hypergraph $\mathcal{H}$, whose node attributes $\mathbf{X}$ are the features that its clustering uses. The auditor holds the audited attribute A and the partition $\mathcal{C}$, and it receives a release of $\mathcal{H}$, a budget B, and any declaration that defines the admissible class, such as a hyperedge size, but not the hyperedges of $\mathcal{H}$. We call an auditor that sees only the release *release-only*. The problem of release determinacy for fairness certification is to (i) decide whether the release determines CGP and HGP and (ii) evaluate the audit and its verdict exactly from the release when it does. When it does not, the problem is to (iii) bound the error of any release-only auditor and (iv) certify the budget conservatively from the release, evaluating HGP exactly from the contingency table for the remaining budgets.

a) Trust model.: The data owner holds the incidence and is trusted to compute every release honestly from one snapshot of it. Certificates rest on this assumption and are audit guarantees, not cryptographic proofs of the computation or privacy guarantees for the release. The pairwise graphs and the random-walk release carry no attribute and are computed by the data owner alone. The auditor can check P in one pass for symmetry, nonnegativity, and row sums equal to the released degrees. An exact audit value, like the per-size counts and the contingency table, depends on the audited attribute of every member of every hyperedge, so it needs attributes and incidence in one place, such as the data owner, the auditor, a trusted third party, or a secure computation between the two. Organizations may not collect or share protected attributes [13], [14], [36], which makes each of these arrangements costly, whereas a certificate needs none.

III. Release Determinacy

In this section, we first derive a linear criterion for the pairwise releases that determine an audit summed over hyperedges. We then apply it to HGP and bound HGP for the releases that do not determine it, where the tightest bound is NP-hard to compute.

A. Determinacy Criterion

Every node-preserving release determines CGP, as CGP depends only on $|C|$, p_g, and $p_{g|C}$ by Eq. (1).

The incidence release and the contingency table determine CGP and HGP, as joining incidence with the attributes and the partition and grouping by (e, g, C) gives every count $n_{e,g,C}$.

The answers for weighted clique expansions and the random-walk release follow from one criterion. We fix the attributes, the partition, and a set $\mathcal{K}$ of sizes, and write $\mathcal{U}_{\mathcal{K}}$ for the node sets with sizes in $\mathcal{K}$. The pair of group and cluster of a node is its *node label*, and the multiset of node labels in a hyperedge is its *label type*.

Definition 2 (Linear pairwise releases and additive audits)**.** A release is a *linear pairwise release* if $\Pi(\mathcal{H}) = \sum_{e\in\mathcal{E}} \zeta(e)$ for nonzero vectors $\zeta(e) \geq 0$ with layers $j = 1, \ldots, J$, each indexed by the unordered node pairs and the nodes, where layer j has the entry $\omega_j(|e|)$ at every unordered pair inside e and $\bar{\omega}_j(|e|)$ at every member of e, for rational coefficient functions ω_j and $\bar{\omega}_j$. An audit is *additive* if $\Lambda(\mathcal{H}) = \sum_{e\in\mathcal{E}} \varphi(e)$, where $\varphi(e)$ is a function of the label type of e.

Weighted clique expansions with rational ω have one layer with $\omega_1 = \omega$, the normalized clique has $\omega_1(k) = 1/(k-1)$, and the random-walk release has $\omega_1(k) = \bar{\omega}_1(k) = 1/k$. The unweighted clique is not a linear pairwise release, and we treat it by explicit counterexamples. The volume and $\mathrm{vol}(\mathcal{H})$ HGP are additive with $\varphi(e) = |e|$ and $|e|\psi_e$, but $\max_e \psi_e$ is not.

Theorem 1 (Determinacy criterion)**.** *Let Π be a linear pairwise release, Λ an additive audit, and n_ι the number of nodes with node label ι. (a) If a vector η satisfies $\varphi(e) = \langle \zeta(e), \eta\rangle$ for every $e \in \mathcal{U}_{\mathcal{K}}$, then $\Lambda(\mathcal{H}) = \langle \Pi(\mathcal{H}), \eta\rangle$ for every input, simple or not. (b) Otherwise, Π does not determine Λ on the class of all multihypergraphs with hyperedges in $\mathcal{U}_{\mathcal{K}}$, as two of them with at most $\min(n_\iota, \max\mathcal{K})$ nodes of each node label ι have the same release and different values of Λ. (c) If such vectors η exist, one of them takes the same value on all pairs with the same two node labels and on all nodes with the same node label. With this vector, (a) is a linear system with one equation for each label type of a size in $\mathcal{K}$ that the counts n_ι admit.*

Proof. Part (a) follows from linearity. For (b), let Ξ be the matrix with the columns $\zeta(e)$ for $e \in \mathcal{U}_{\mathcal{K}}$. If no η exists, φ lies outside the row space of the rational matrix Ξ, so some integral z with $\Xi z = 0$ has $\langle \varphi, z\rangle \neq 0$. The multihypergraphs with multiplicities z^+ and z^- have equal releases, both are nonempty given that the columns of Ξ are nonnegative and nonzero, and their values of Λ differ by $\langle \varphi, z\rangle$. For (c), a permutation of nodes with equal node labels maps solutions to solutions, so the average of a solution over these permutations has the stated form. Its equation for a hyperedge depends on the label type, and types with at most $\max\mathcal{K}$ members occur, so capping every n_ι at $\max\mathcal{K}$ leaves the system unchanged, and (b) applies to the capped node set. □

In words, on multihypergraphs a release determines an additive audit exactly when the contribution of each hyperedge to the audit is one fixed linear function of its contribution to the release. This criterion follows the span condition of SUM-query auditing [37], [38], like the linear criterion of query determinacy under bag semantics [39]. Our new results are its instances for HGP, the separation between simple hypergraphs and multihypergraphs, and the hardness of the tightest certificate.

Theorem 2 (Simple hypergraphs)**.** *Let Π and Λ be as in Theorem 1. (a) On simple hypergraphs, Π determines Λ exactly when $\langle \varphi, z\rangle = 0$ for every $z \in \{0, \pm 1\}^{\mathcal{U}_{\mathcal{K}}}$ with $\sum_e z_e\zeta(e) = 0$. The condition of Theorem 1(a) is sufficient for every $\mathcal{K}$ and necessary whenever these vectors z span all rational solutions of this equation. (b) The vectors z of (a) span all rational solutions if every hyperedge has the declared uniform size r and some $\omega_j(r) \neq 0$. (c) If the condition of Theorem 1(a) then fails, then $r \geq 3$, and two simple hypergraphs with four hyperedges each on $r+3$ nodes, at most $\min(n_\iota, r)$ of them with node label ι, have the same release and different values of Λ. No two distinct inputs on fewer than $r+3$ nodes have the same release.*

Proof. Two simple inputs have the same release exactly when their difference z has entries in $\{0, \pm 1\}$ and $\Xi z = 0$, which gives the first claim of (a), and the sufficiency of the condition follows from Theorem 1(a). For (b) and (c) and a single size r, every r-set that contains a node v contains $r-1$ pairs at v, so $(r-1)(W_1 z)_v = \sum_{u\neq v}(W_2 z)_{uv}$ for the inclusion matrices W_1 and W_2 of nodes and of pairs versus r-sets, and the kernel of Ξ is that of W_2. This kernel is zero on at most $r+2$ nodes by the rank of W_2 [40], [41], applied to complements when there are $r+1$ nodes, and it is generated by the elementary trades [42], [43]. An elementary trade $(x_1 - y_1)(x_2 - y_2)(x_3 - y_3)S$ on $r+3$ nodes is the sum of the hyperedges $\{w_1, w_2, w_3\} \cup S$ over $w_i \in \{x_i, y_i\}$, signed by the parity of the number of y_i, for a set S of $r-3$ further nodes, and it has four hyperedges on each side. A pair meets at most two of the three factors, so the signed hyperedges that contain it cancel over the remaining factor, and every pair count of an elementary trade is zero. If the condition of Theorem 1(a) fails, it fails on the capped node set by Theorem 1(c), which then has at least $r+3$ nodes, and one of its elementary trades has $\langle \varphi, z\rangle \neq 0$. □

The spanning condition of Theorem 2(a) holds for a declared uniform size by part (b) but can fail for mixed sizes.

Remark 3 (Normalized audits)**.** If Π determines vol, then it determines Λ/vol exactly when it determines Λ, as inputs with the same release have the same positive volume.

B. Instances for Weighted Cliques and the Random-Walk Release

Weighted cliques of a declared uniform size and the random-walk release determine HGP for hyperedges of up to three members and not beyond, and the crossing pairs of a hyperedge explain this boundary. Call a pair of nodes *crossing* if its members differ in both group and cluster, write $\chi_{uv} = \mathbf{1}[A(u) \neq A(v) \wedge \mathcal{C}(u) \neq \mathcal{C}(v)]$ for its indicator, and

let N_e sum χ_{uv} over the pairs of members of a hyperedge e. Let $U_P = \sum_{u,v} P_{uv}\chi_{uv} / \sum_{u,v} P_{uv}$ be the share of the mass of the random-walk release P on ordered crossing pairs, where the sums run over all ordered pairs of nodes, including $u = v$.

Lemma 1 (Crossing pairs)**.** *For every hyperedge e, $\psi_e \le 2N_e/|e|^2$, with equality if $|e| \le 3$ or if, inside e, the cluster of each member is a function of its group or its group is a function of its cluster.*

Proof. Let $k = |e|$. For a group g and a cluster C, let $n_{11} = n_{e,g,C}$, and let n_{10}, n_{01}, and n_{00} count the members of e that are in g but outside C, in C but outside g, and in neither. By Eq. (3), $\psi_e = \sum_{g,C} |R_{gC}|/(2k^2)$ with the residual $R_{gC} = kn_{11} - (n_{11}+n_{10})(n_{11}+n_{01}) = n_{11}n_{00} - n_{10}n_{01}$, and the residuals sum to 0, as the products $(n_{11}+n_{10})(n_{11}+n_{01})$ sum to k^2 over the cells, and $R_{gC} = -n_{10}n_{01} \le 0$ if $n_{11} = 0$, so $\sum_{g,C} |R_{gC}| = 2\sum_{g,C} \max(R_{gC}, 0) \le 2\sum_{g,C} n_{11}n_{00} = 4N_e$, as a member in group g and cluster C differs in both group and cluster from exactly n_{00} members, so $\sum_{g,C} n_{11}n_{00} = 2N_e$ counts each crossing pair twice. Each term is tight when $n_{11}n_{00} = 0$ or $n_{10}n_{01} = 0$, which holds for every cell if the clusters inside e are a function of the groups or conversely, and also if $k \le 3$, as a cell with $n_{11}n_{00} > 0$ and $n_{10}n_{01} > 0$ needs four members. □

The bound can be strict. The hyperedge $\{a_1, a_2, b_1, b_2\}$ with groups $\{a_1, a_2\}$ and $\{b_1, b_2\}$ and clusters $\{a_1, b_1\}$ and $\{a_2, b_2\}$ spreads each group over the clusters like the whole hyperedge, so $\psi_e = 0$, but its crossing pairs a_1b_2 and a_2b_1 give $2N_e/|e|^2 = 1/4$. A gap between this bound and ψ_e needs at least four members.

Corollary 1 (Uniform weighted cliques)**.** *Assume every hyperedge has the declared uniform size $r \ge 2$ and use exact positive size-only weights $\omega(r)$. (i) For $r \le 3$ and any fixed attributes and partition, the weighted clique determines HGP, simple or not, with* $\mathrm{HGP} = \big((r-1)/r\big) \sum_{uv} w_{uv}\chi_{uv} / \sum_{uv} w_{uv}$*, where $\sum_{uv}$ runs over unordered pairs of distinct nodes. (ii) For every $r \ge 4$ there are attributes, a partition, and two simple hypergraphs with identical weighted cliques whose HGP values are $(r-4)/r^2$ and $(r-2)/r^2$.*

Proof. (i) The weighted clique with weight $\omega(r)$ is $\omega(r)$ times the one with weight 1, and the formula is invariant under this scaling, so we may take $\omega(r) = 1$. Every hyperedge has $\psi_e = 2N_e/r^2$ by Lemma 1, so $\eta_{uv} = 2\chi_{uv}/r$ satisfies Theorem 1(a) for $\Lambda = \mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$, as each pair of a hyperedge carries the weight 1, yielding $\langle \zeta(e), \eta \rangle = 2N_e/r = r\psi_e$. With m hyperedges, $\sum_{uv} w_{uv} = m\binom{r}{2}$ and $\mathrm{vol}(\mathcal{H}) = rm = 2\sum_{uv} w_{uv}/(r-1)$, which yields the formula.

(ii) Take nodes $0, \ldots, 6$ with group $a = \{0, 2, 3\}$ and cluster $C_1 = \{0, 4, 5\}$, the other nodes forming group b and cluster C_2, and a set Z of $r-4$ further nodes of group b and cluster C_2. The elementary trade $(0-1)(2-5)(4-3)(\{6\} \cup Z)$ of Theorem 2 has the hyperedges $\{024, 035, 123, 145\}$ on its positive side and $\{023, 045, 124, 135\}$ on its negative side, each extended by $\{6\} \cup Z$, so the two sides share every linear pairwise release, including the pair multiplicities by hyperedge size and the node degrees. At $r = 4$, every hyperedge of the first side lies in one cluster, has one group, or splits each of its groups evenly over the clusters, so $\psi_e = 0$ by Eq. (3), and every hyperedge of the second side has three members of one group, split one to two over the clusters, and one member of the other group in the cluster that holds two of them, so $\psi_e = 1/8$. For $r > 4$, the members of Z join every hyperedge, yielding $(r-4)/r^2$ and $(r-2)/r^2$. □

Corollary 2 (Random-walk release)**.** *(i) If no hyperedge has more than three members, the random-walk release determines HGP, also for mixed sizes and multihypergraphs, and* $\mathrm{HGP} = U_P$*. (ii) If the admissible sizes include some $r \ge 4$, no release computed from pair multiplicities by hyperedge size, which fix the node degrees, determines HGP, even with mixed sizes.*

Proof. (i) The vector $\eta = 2\chi$ on pairs and $\eta = 0$ on nodes gives $\langle \zeta(e), \eta \rangle = 2N_e/|e| = |e|\psi_e$ for every admissible hyperedge by Lemma 1, and $\mathrm{vol}(\mathcal{H}) = \sum_e \sum_{u,v \in e} 1/|e|$ is the sum of all entries of P. With repeated hyperedges of two and three members, three copies of a triple and two copies of each of its pairs have the same off-diagonal entries of P and the same $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$. Their diagonal entries are 1 and 2 and their volumes 9 and 12, so the diagonal fixes the volume. (ii) The two inputs of Corollary 1(ii) use hyperedges of size r and agree on these multiplicities, and their HGP values differ. □

Part (ii) applies to the random-walk release and to every positive size-only weight rule.

Corollary 3 (Hyperedges of two and three members)**.** *Let the admissible sizes be $\{2, 3\}$ and let some admissible triple f contain $N_f > 0$ crossing pairs. (i) On multihypergraphs, a weighted clique expansion with positive rational $\omega(2)$ and $\omega(3)$ determines* $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$ *exactly when $\omega(3)/\omega(2) = 2/3$ and* $\mathrm{vol}(\mathcal{H})$ *exactly when $\omega(3)/\omega(2) = 1/2$, and it never determines HGP. (ii) On simple hypergraphs whose hyperedges are not declared disjoint, the unweighted clique does not determine HGP, and on four or more nodes neither does the normalized clique.*

Proof sketch. For (i), we solve the system of Theorem 1(a) on hyperedges of two and three members. For $\omega(3)/\omega(2) = p/q$ in lowest terms, q copies of f and p copies of each of its pairs share the weighted clique and have HGP $2N_f/9$ and $N_f/6$, which are $4/9$ and $1/3$ for the running example. For (ii), with nodes $0, 1, 2$ in group a and cluster C_1 and node 3 in group b and cluster C_2, the inputs $\{03, 13, 23, 012\}$ and $\{013, 023, 123\}$ share the normalized clique and have HGP $1/3$ and $4/9$. The complete proof appears in the extended version [44]. □

Corollary 4 (Simple hyperedges of two and three members)**.** *On simple hypergraphs with admissible sizes $\{2, 3\}$, a weighted clique expansion with $\omega(3)/\omega(2) = p/q$ in lowest terms and $p \ge 2$ determines HGP for every attribute and partition. This includes the off-diagonal of the random-walk release, whose ratio is $2/3$.*

TABLE II

DETERMINACY OF HGP BY RELEASE. A RELEASE DETERMINES HGP (✓) IF ALL ADMISSIBLE INPUTS THAT SHARE IT HAVE THE SAME HGP FOR EVERY ATTRIBUTE AND PARTITION, AND OTHERWISE IT DOES NOT (✗). WEIGHTS ARE EXACT, HYPEREDGE SIZES ARE DECLARED, AND RATIOS p/q ARE IN LOWEST TERMS.

Release	Admissible class	HGP	Support
Incidence, contingency table	all	✓	§III-A
Weighted clique, $\omega(r) > 0$	uniform size 2 or 3	✓	Cor. 1(i)
Random-walk release	size ≤ 3	✓	Cor. 2(i)
Unweighted clique or P	declared disjoint	✓	§III-B
Weighted clique, $\omega(3)/\omega(2) = p/q$, $p \geq 2$	simple, sizes $\{2, 3\}$	✓	Cor. 4
Weighted clique or P	some size ≥ 4	✗	Cor. 1(ii), 2(ii)
Weighted clique, rational $\omega > 0$	multi, sizes $\{2, 3\}$	✗	Cor. 3(i)
Normalized clique	simple, sizes $\{2, 3\}$, ≥ 4 nodes	✗	Cor. 3(ii)
Unweighted clique	not declared disjoint	✗	Cor. 3(ii)
Attribute-similarity graph	–	✗	Fig. 1(c)

Proof. For two simple inputs with the same release, let $c \in \{0, \pm 1\}$ and $t \in \mathbb{Z}$ be the differences of their hyperedges of two members at a pair and of their triples that contain it. Then $\omega(2)c + \omega(3)t = 0$ gives $qc = -pt$, so p divides c, and $c = t = 0$ at every pair. The two inputs share their hyperedges of two members, and their triples differ by a vector with zero pair counts, on which $\mathrm{vol}(\mathcal{H})$ and $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$ do not change by the proof of Corollary 1(i). □

A release computed without incidence, such as an attribute-similarity graph, is the same for all inputs with the given attributes, so it determines only the audits that are constant on the class. HGP is not constant on the class, as the running example shows.

Table II summarizes these results. If the data owner declares the hyperedges disjoint, the unweighted clique and P also determine HGP, as the connected components of the clique, or of the nonzero entries of P, are then the hyperedges.

For other additive audits, the boundary differs. Take two groups, two clusters, and the 256 choices of zero to three nodes for each of the four node labels. For 129 of these choices, the 3-uniform weighted clique leaves the squared dependence $\varphi(e) = |e| \sum_{g,C} (p_e(g, C) - p_e(g)\, p_e(C))^2$ and the membership-weighted mutual information between group and cluster undetermined by Theorem 1(c), although it determines HGP for all of them. The boundary between hyperedge sizes three and four belongs to the total-variation form, which is the dissimilarity family of segregation indexes in wide use since Duncan and Duncan [45]. The worst gap $\max_{e,g} \mathrm{TV}(q_{e,g}, q_e)$ is not additive. The two inputs of Corollary 1(ii) at $r = 4$, which share the weighted clique and the random-walk release, have the worst gap 0 and $1/4$, so these releases leave it undetermined once hyperedges of four members are admissible.

C. Bounds on Auditor Error

For a fixed release, let $h_{\min}$ and $h_{\max}$ be the infimum and supremum of HGP over the compatible inputs, which are the hypergraphs that produce the release with the same attributes and partition. We call $[h_{\min}, h_{\max}]$ the compatible range, as in the range semantics of aggregate queries over inconsistent databases [46], [47]. Every release-only auditor returns one value f for all compatible inputs, so its worst-case absolute error $\max(|f - h_{\min}|, |f - h_{\max}|)$ is at least $(h_{\max} - h_{\min})/2$, and the midpoint attains this bound. This includes an auditor that evaluates HGP on a hypergraph reconstructed from the pairwise graph [19].

Theorem 3 (Parameterized gap)**.** *For every $r \geq 3$ there are two simple hypergraphs on r nodes with identical unweighted cliques K_r, attributes, and partition whose HGP values differ by $\gamma(r) = \lceil r/2 \rceil \lfloor r/2 \rfloor (r-2)/(r^2(r-1))$. The value $\gamma(r) \to 1/4$ as $r \to \infty$, with $\gamma(3) = 1/9$, $\gamma(4) = 1/6$, and $\gamma(32) \approx 0.242$.*

Proof. Split the r nodes into blocks of $s = \lceil r/2 \rceil$ and $t = \lfloor r/2 \rfloor$ nodes, where the first forms group a and cluster C_1 and the second forms group b and cluster C_2. One hyperedge with all r nodes has $\mathrm{HGP} = 2st/r^2$, as $\mathrm{TV}(q_{e,a}, q_e) = t/r$ and $\mathrm{TV}(q_{e,b}, q_e) = s/r$, and the $\binom{r}{2}$ pairs as hyperedges have $\mathrm{HGP} = st/(r(r-1))$, as each of the st cross-group pairs has $\psi = 1/2$ and each same-group pair has $\psi = 0$. Both have the unweighted clique K_r, and the values differ by $\gamma(r)$. □

The running example is the case $r = 3$.

In general, the dyadic surrogate is the HGP of the hypergraph with one size-2 hyperedge per pair of the clique. This hypergraph is compatible with the unweighted clique unless the data owner declares a minimum size above two, which excludes it, or declares the hyperedges disjoint, which makes the connected components of the clique the hyperedges. If it is compatible, the gap between $\mathrm{HGP}_{\mathrm{true}}$ and $\mathrm{HGP}_{\mathrm{sur}}$ lower-bounds the width of the compatible range, so half of the gap lower-bounds the worst-case error of any release-only auditor. HGP remains undetermined for some attributes and partitions if hyperedges of four or more members are admissible, by Corollaries 1(ii) and 2(ii), whose paired inputs share the number and sizes of hyperedges and the node degrees.

D. Conservative Certificates and Their Hardness

Certifying a budget requires an upper bound on HGP, which the random-walk release and the normalized clique supply together with the attributes and the partition. Let U_Q be the weighted share of crossing pairs in the normalized clique,

$$U_Q = \frac{\sum_{uv} w_{uv}\, \chi_{uv}}{\sum_{uv} w_{uv}},$$

where the sums run over unordered pairs of distinct nodes, and write $U_{Q,\bar{r}} = \big((\bar{r} - 1)/\bar{r}\big)\, U_Q$ for a cap $\bar{r}$ on the hyperedge size.

Proposition 2 (Conservative certificates)**.** *(a)* $\mathrm{HGP} \leq U_P$*, with equality if every hyperedge has at most three members*

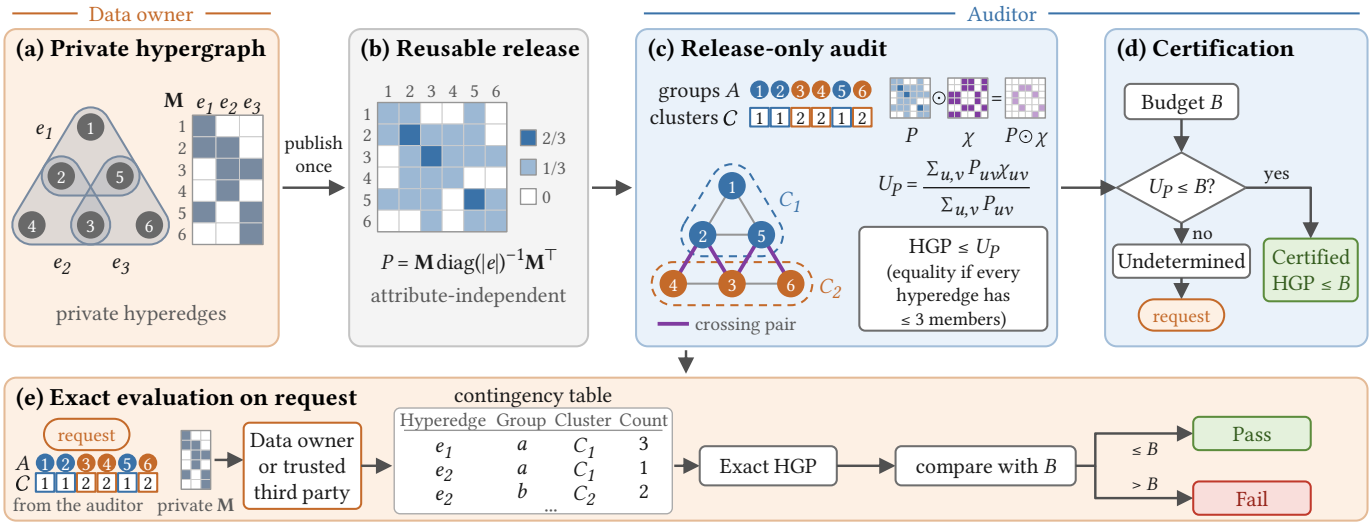


Fig. 2. Certify-then-request. The auditor certifies B from the release if $U_P \le B$ and otherwise requests exact counts.

or if, inside every hyperedge, the cluster of each member is a function of its group or its group is a function of its cluster. (b) $U_P \le U_Q$, and $U_P \le U_{Q,\bar r}$ if every hyperedge has at most $\bar r$ members, so U_Q and $U_{Q,\bar r}$ also bound HGP. Each bound takes one pass over its release.

Proof. The mass of P is $\sum_e \sum_{u,v\in e} 1/|e| = \mathrm{vol}(\mathcal{H})$, and each member pair of e with $\chi_{uv} = 1$ appears twice with weight $1/|e|$, and no diagonal entry has $\chi_{uu} = 1$, so $U_P = \big(2/\mathrm{vol}(\mathcal{H})\big)\sum_e N_e/|e|$. By Lemma 1, weighting the bound on ψ_e by $|e|/\mathrm{vol}(\mathcal{H})$ and summing gives (a). For (b), exchanging the order of summation gives $U_Q = \big(2/\mathrm{vol}(\mathcal{H})\big)\sum_e N_e/(|e|-1)$, as the weights of the normalized clique sum to $\sum_e \binom{|e|}{2}/(|e|-1) = \mathrm{vol}(\mathcal{H})/2$. Finally, $1/|e| < 1/(|e|-1)$ gives $U_P \le U_Q$, and $1/|e| = \big(1/(|e|-1)\big)\big((|e|-1)/|e|\big) \le \big(1/(|e|-1)\big)\big((\bar r-1)/\bar r\big)$ for $|e| \le \bar r$ gives $U_P \le U_{Q,\bar r}$. □

The bound U_P is computed from P, or from T with the degrees d_v, as $P = \mathrm{diag}(d)\,T$. It is tight, as $U_P = \mathrm{HGP}$ on H_1 and H_2 by part (a), so no bound κU_P with $\kappa < 1$ holds. On an exact release, U_P needs no cap on hyperedge size.

The normalized-clique surrogate evaluates Eq. (2) on one size-2 hyperedge per released pair, weighted by w_{uv}. A size-2 hyperedge has $\psi = 1/2$ when its members differ in both group and cluster and $\psi = 0$ otherwise, so the surrogate equals $U_Q/2$.

On the scale of $\bar D$, a certificate also needs a lower bound on $\bar I$, which the exact random-walk release supplies exactly and a rounded or pruned one bounds from below. Let $\xi_{uv} = \mathbf{1}[A(u) \ne A(v)]$ indicate a cross-group pair.

Proposition 3 (Certificates on the dissimilarity scale)**.** *(a) $\bar I = \sum_{u,v} P_{uv}\xi_{uv} \big/ \sum_{u,v} P_{uv}$ over all ordered pairs, so $\bar D \le U_P/\bar I = \sum_{u,v} P_{uv}\chi_{uv} \big/ \sum_{u,v} P_{uv}\xi_{uv}$, which is 0 if $\bar I = 0$. (b) If every hyperedge has at least $r_{\min}$ and at most $\bar r$ members, then $\bar D \le c \sum_{uv} w_{uv}\chi_{uv} \big/ \sum_{uv} w_{uv}\xi_{uv}$ for the normalized clique with $c = \big((\bar r-1)/\bar r\big)\big(r_{\min}/(r_{\min}-1)\big)$, and $c = 2$ holds without any declaration. No smaller constant holds in (a) or (b).*

Proof. (a) Each hyperedge e adds $1/|e|$ to its $|e|^2$ ordered member pairs, $|e|^2 I_e$ of them across groups, so $\sum_{u,v} P_{uv} = \mathrm{vol}(\mathcal{H})$ and $\sum_{u,v} P_{uv}\xi_{uv} = \sum_e |e|\, I_e = \mathrm{vol}(\mathcal{H})\,\bar I$, and $\mathrm{HGP} \le U_P$ by Proposition 2(a), with equality on H_1. (b) A hyperedge e adds $N_e/(|e|-1)$ to $\sum_{uv} w_{uv}\chi_{uv}$ and $|e|^2 I_e/(2|e|-2)$ to $\sum_{uv} w_{uv}\xi_{uv}$, whereas $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP} \le \sum_e 2N_e/|e|$ by Lemma 1 and $\mathrm{vol}(\mathcal{H})\,\bar I = \sum_e |e|\, I_e$. As $(k-1)/k$ increases in k, $\sum_e 2N_e/|e| \le 2\frac{\bar r-1}{\bar r}\sum_{uv} w_{uv}\chi_{uv}$ and $\sum_e |e|\, I_e \ge 2\frac{r_{\min}-1}{r_{\min}}\sum_{uv} w_{uv}\xi_{uv}$, and dividing gives (b). For the constants, one hyperedge of $\bar r$ members whose groups are its clusters, together with k hyperedges of $r_{\min}$ members that lie in one cluster and hold two groups, gives a ratio of the two sides that tends to c as k grows, and c tends to 2 as $\bar r$ grows with $r_{\min} = 2$. □

Under the declared sizes of our datasets, c is 1.96 for hyperedges of at most 50 members and 1.47 for hyperedges of 3 to 50 members.

Proposition 4 (Certificates as dual solutions)**.** *Let Π be a linear pairwise release and Λ an additive audit. (a) Every vector η with $\varphi(e) \le \langle \zeta(e), \eta\rangle$ for all $e \in \mathcal{U}_K$ gives the conservative certificate $\Lambda(\mathcal{H}) \le \langle \Pi(\mathcal{H}), \eta\rangle$ for every compatible input, simple or not. (b) For $\Lambda = \mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$, the vector $\eta = 2\chi$ on pairs and $\eta = 0$ on nodes is feasible for the random-walk release and yields U_P after division by the volume. For the normalized clique, $\eta = 2\chi$ yields U_Q, and $\eta = 2\big((\bar r-1)/\bar r\big)\chi$ yields $U_{Q,\bar r}$ if no admissible size exceeds $\bar r$. (c) For a release that fixes the volume, the smallest certificate of this form for $\Lambda = \mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$, divided by the volume, is the value of the linear relaxation of the largest HGP over the compatible multihypergraphs.*

Proof. The first claim follows from $x \geq 0$ for the multiplicities x. Feasibility is the inequality $\psi_e \leq 2N_e/|e|^2$ of Lemma 1, which yields $\langle \zeta(e), 2\chi \rangle = 2N_e/|e| \geq |e|\psi_e$ for P, and $2N_e/(|e|-1) \geq 2N_e/|e|$ for the normalized clique, where $(\bar{r}-1)/\bar{r} \geq (|e|-1)/|e|$ for $|e| \leq \bar{r}$. The relaxation is feasible, as the input is a solution, and bounded, given that every column has positive mass, so linear-programming duality makes its value equal to the smallest such certificate. □

Released entries are often rounded or pruned, as in the cond-mat file and the operator of AHRC. The auditor receives an interval $[\ell_o, h_o] \subseteq [0, \infty)$ that contains the exact value of each listed entry o, an ordered pair of nodes for P, possibly equal, and an unordered pair of distinct nodes for the normalized clique. For the symmetric P, the auditor also lists the transpose of each listed entry and gives both the intersection of their intervals, an unlisted entry having the interval $[0, \infty)$. Let N_h sum h_o over the listed crossing entries and M_ℓ sum ℓ_o over the other listed entries. The bounds below count every crossing entry at its upper end and every other entry at its lower end, so they stay conservative.

Proposition 5 (Certificates from inexact releases)**.** *(a) If every entry of positive value is listed, then* $\mathrm{HGP} \leq U_P \leq N_h/(N_h + M_\ell)$ *for* P, $\mathrm{HGP} \leq U_Q \leq N_h/(N_h + M_\ell)$ *for the normalized clique, and* $\mathrm{HGP} \leq U_{Q,\bar{r}} \leq \big((\bar{r}-1)/\bar{r}\big)N_h/(N_h + M_\ell)$ *if no hyperedge has more than* $\bar{r}$ *members, where a ratio with* $N_h = 0$ *is zero. (b) If the degrees* d_v *are released, then* $\mathrm{HGP} \leq U_P \leq 1 - M_\ell/\mathrm{vol}$ *with* $\mathrm{vol} = \sum_v d_v$*, whichever entries of* P *are missing. (c) Under (b), if the sum* L_ℓ *of* ℓ_o *over the listed cross-group entries is positive, then* $\bar{I} \geq L_\ell/\mathrm{vol}$ *and* $\bar{D} \leq (\mathrm{vol} - M_\ell)/L_\ell$*.*

Proof. (a) As every entry of positive value is listed, $U_P = N/(N+M)$ for the crossing mass $N \leq N_h$ and the other mass $M \geq M_\ell$, and $x/(x+y)$ increases in x and decreases in y for $x, y \geq 0$. The weights of the normalized clique behave alike, and the stated bounds on HGP follow from Proposition 2. (b) Row v of P sums to d_v, so P has the mass vol, and its non-crossing mass is at least M_ℓ, as unlisted entries are nonnegative, which gives $U_P \leq 1 - M_\ell/\mathrm{vol}$. (c) Likewise, $\mathrm{vol}\,\bar{I} \geq L_\ell$ by Proposition 3(a), and dividing (b) by this bound gives (c). □

Part (a) applies under relative error and under an absolute radius below $1/\bar{r}$ for P or $1/(\bar{r}-1)$ for the normalized clique, as no positive entry is smaller. Six significant digits, as in the cond-mat file, give a relative radius of at most $5 \cdot 10^{-6}$. Double precision gives T a relative error of at most $k2^{-53}/(1-k2^{-53})$ for $k = d_{\max} + 2$ [48], which is at most $1.05 \cdot 10^{-13}$ on our datasets. Part (b) applies to rows pruned without rescaling, as in AHRC, and to T rounded to p decimals with zero entries dropped, as a positive entry in row u of T is at least $1/(\bar{r}d_u)$.

A release-only auditor certifies a budget B exactly when $h_{\max} \leq B$, so $h_{\max}$ is the tightest certificate. Unlike the relaxation of Proposition 4, $h_{\max}$ is NP-hard to compute or to bound closely from above.

Theorem 4 (Hardness of the tightest certificate)**.** *Given a release, an attribute with two groups, a partition into two clusters, a rational budget* B*, and one compatible input, deciding whether* $h_{\max} > B$ *is NP-complete in each setting below. Unless* $\mathrm{P} = \mathrm{NP}$*, no polynomial-time auditor returns on every input a bound* U *with* $h_{\max} \leq U < h_{\max} + \delta$*. (a) For a fixed* $r \geq 4$*, the inputs are simple and* r*-uniform, the release is an exact weighted clique expansion with rational* $\omega(r) > 0$ *or the random-walk release* P*, and* $\delta = 8/(r^2 m)$ *for the number* m *of hyperedges. Deciding whether* $h_{\max} = U_P$ *is NP-complete as well. (b) For a fixed* $R \geq 3$*, the inputs are simple hypergraphs or multihypergraphs whose hyperedges have two to* R *members, the release is a weighted clique expansion with* $\omega(2) = \cdots = \omega(R)$*, and* $\delta = 1/(6M)$ *for* M *released pairs.*

Proof. The release fixes the multiplicity μ_{uv} of every pair, which bounds the multiplicity of every hyperedge, so the compatible inputs are finitely many, and one with HGP above B, checked exactly, certifies $h_{\max} > B$ in NP, also for $h_{\max} = U_P$ as $\mathrm{HGP} \leq U_P$. We prove part (b) in full and outline part (a).

Part (b). We reduce from partitioning the edges of a graph $G = (V_G, E_G)$ into copies of K_R [49], with $M' = |E_G| \geq 1$, as an edgeless graph maps to a fixed yes-instance. We place V_G in group a and C_1, add M' disjoint pairs $x_i y_i$ with x_i in group a and C_1 and y_i in group b and C_2, and release $\omega(2)$ on every edge of G and pair $x_i y_i$. As $\omega(k) = \omega(2)$ for $k \leq R$, every released pair lies in exactly one hyperedge and no other pair lies in a hyperedge. The compatible inputs consist of the pairs $x_i y_i$ and of cliques of two to R nodes of G that partition E_G, and the input with one size-2 hyperedge per pair is among them. Hyperedges inside V_G have $\psi_e = 0$ and each pair $x_i y_i$ has $\psi_e = 1/2$, so $\mathrm{HGP} = M'/(\mathrm{vol}_G + 2M')$ with $\mathrm{vol}_G = \frac{2M'}{R-1} + \sum_e \frac{|e|(R-|e|)}{R-1}$ over the cliques, where a term vanishes if $|e| = R$ and is at least one otherwise, as $(|e|-1)(R-|e|-1) \geq 0$. With $S^* = 2M'R/(R-1)$, which satisfies $S^* \leq 3M'$ and $S^* + 1 \leq 4M'$, we have $h_{\max} = M'/S^*$ if G has a K_R-partition and $h_{\max} \leq M'/(S^*+1)$ otherwise. The budget $B = 2M'/(2S^*+1)$ separates the cases, and the gap $M'/(S^*(S^*+1))$ is at least $1/(12M') = 1/(6M)$ for the $M = 2M'$ released pairs, so a bound U with $h_{\max} \leq U < h_{\max} + 1/(6M)$ satisfies $U \geq M'/S^*$ exactly when G has a K_R-partition.

Part (a). We reduce from completing a partial Latin square [50], which is possible exactly when the edges of its defect graph G partition into triangles. Every edge of G becomes a hyperedge of four members. Switch and absorber gadgets, families of 4-member hyperedges that cover the same pairs as an alternative family, let a compatible input replace the hyperedges of the three edges of a triangle uvw by one hyperedge $uvwz$, for any set of edge-disjoint triangles. Conversely, the given input covers every pair at most once, so a compatible input partitions the released pairs into 4-cliques, and the only such 4-cliques are the hyperedges of the gadget families. Groups and clusters make $U_P - \mathrm{HGP}$ the same positive amount on each hyperedge of an edge at a row

and zero on every other hyperedge, and a hyperedge $uvwz$ covers two such edges, so $h_{\max} = U_P$ exactly when the square can be completed. Padding every hyperedge with $r-4$ common nodes gives $r \geq 5$. On the smallest instance, a square of order one with its cell empty, exhaustive enumeration over the 37 nodes and 150 released pairs finds exactly the 53 hyperedges of the gadget families as 4-cliques. The complete construction and its verification are in the supplementary material [51]. □

In the settings of Theorem 4, deciding $h_{\max} \leq B$ is the complement of an NP-complete problem and hence coNP-complete. At a uniform size $r \in \{2, 3\}$, the release fixes $h_{\max}$, by Corollaries 1(i) and 2(i), so computing the tightest certificate takes polynomial time there. In part (a), the release fixes m together with U_P. A gap δ that depends on r alone, not on m, also holds in part (a), by a reduction from a gap version of the independence number in graphs of maximum degree three [52], as we prove in Appendix A. For bounded hyperedge sizes, the relaxation of Proposition 4 gives the tightest dual certificate in polynomial time.

IV. Exact Evaluation from Contingency Tables

If a conservative certificate leaves a budget undetermined, the contingency table supplies sufficient statistics for exact evaluation, and so do the per-size counts and sums derived from it. In certify-then-request, the data owner or a trusted third party computes the table and these counts, so the auditor never receives the hyperedges, and the attributes leave the auditor only for the budgets that no certificate settles. In this section, we obtain the contingency table, evaluate HGP from it, and combine it with the certificates, as summarized in Figure 2.

a) Obtaining the contingency table.: A party that holds the incidence and the attributes, such as a trusted third party, can release the contingency table directly, as a table with the fields `hyperedge_id`, `attr`, `group_val`, `cluster_id`, and `n`. If the auditor holds the incidence release, it joins the release with the attributes and the partition and groups by hyperedge, group, and cluster.

b) Sparse evaluation.: Let $n_{e,g} = \sum_C n_{e,g,C}$ and $n_{e,C} = \sum_g n_{e,g,C}$. By the proof of Lemma 1, the residuals $|e|\, n_{e,g,C} - n_{e,g} n_{e,C}$ sum to zero and are nonpositive when $n_{e,g,C} = 0$, and Eq. (3) becomes

$$\psi_e = \frac{1}{|e|^2} \sum_{(g,C):\, n_{e,g,C} > 0} \max\big(|e|\, n_{e,g,C} - n_{e,g} n_{e,C},\, 0\big). \quad (4)$$

Every residual is an integer, so HGP is an exact rational. In H_1 of Figure 1, the hyperedge $\{u, v, w\}$ has the nonzero counts $n_{e,a,C_1} = 2$ and $n_{e,b,C_2} = 1$, both residuals equal 2, and $\psi_e = 4/9$. In H_2, $\{u, v\}$ has $\psi_e = 0$ and the others have $\psi_e = 1/2$, so HGP $= 1/3$.

c) Per-size counts.: An SQL query takes the grouped counts $n_{e,g,C}$ and sums them by hyperedge, by hyperedge and group, and by hyperedge and cluster to obtain $|e|$, $n_{e,g}$, and $n_{e,C}$. It then groups by $|e|$ to return the number of hyperedges and the integer sum of $|e|^2 \psi_e$ for each hyperedge size. HGP follows from these per-size counts and sums, two integers per attribute and occurring hyperedge size, so an auditor that needs the verdict alone can request them in place of the contingency table. Nodes with a missing attribute value form one additional group, so the query counts every member of a hyperedge in $|e|$. After aggregation, HGP evaluation requires $O\big(\sum_A |\Psi_A|\big)$ operations for $|\Psi_A|$ nonzero rows of Ψ_A.

Algorithm 1 Certify-then-request for one attribute

Require: release Π, attribute A, partition $\mathcal{C}$, budget B, declared disjointness, cap $\bar{r}$, or entry intervals, if any

1: **if** exact Π determines HGP (Table II) **then return** its verdict
2: $U \leftarrow$ the smallest bound of Propositions 2 and 5 for Π, or $+\infty$
3: **if** $U \leq B$ **then return** pass ▷ certified, no request
4: send A and $\mathcal{C}$ to the data owner or a trusted third party
5: receive the per-size counts and sums of Eq. (4)
6: **return** pass if the exact HGP is at most B, and fail otherwise

d) Certify-then-request.: The decision rule for one audited attribute is listed in Algorithm 1 and shown in Figure 2.A budget that a bound of Propositions 2 and 5 certifies also passes exact evaluation, provided that no hyperedge exceeds the cap $\bar{r}$ of $U_{Q,\bar{r}}$. For example, an auditor that holds T with degrees from an AHRC or AHCKA pipeline, or P from the data owner, certifies every budget of at least U_P without a request. For a lower budget, the data owner or a trusted third party returns the per-size counts and sums, or the contingency table to localize violations.

V. Experiments

In this section, we answer two research questions. (RQ1) How much information do pairwise releases lose for HGP verdicts? (RQ2) How many budgets can a held release certify without joint access to the attributes and the hyperedges?

A. Setup

We audit the partitions of five clustering methods on six real datasets and compare auditors on the same partition.

a) Datasets.: We use six real hypergraph datasets, summarized in Table III, which are MovieLens-1M (ML-1M) [53], OpenAlex-DB (OA-DB) and OpenAlex-CS-3+ (OA-CS-3+) [54], Pokec-Hobby (Pokec) [55], IMDb-Cast [56], and ACS-PUMS households (ACS-PUMS-HH) [57]. An ML-1M hyperedge gathers the users who rated one film with four or more stars in one calendar month, and a Pokec hyperedge the users who list one normalized hobby token. Each type of hyperedge matches a decision made per cluster, such as outreach programs for households, invitations for author teams, and recommendation slates for film casts and for the users of one hyperedge. Outside ACS-PUMS-HH, every hypergraph retains hyperedges of 3 to 50 members, or 3 to 25 for OA-CS-3+, so every method and audit sees the same hyperedges. The lower end excludes hyperedges of two members, which a pairwise release represents exactly by Corollary 1(i). On

TABLE III
DATASETS. † MARKS ATTRIBUTES OUTSIDE THE MAIN EXPERIMENTS AND ‡ AN UNTRAINED CAHC. CE+FL IS CE+FAIR-LOUVAIN.

				Number of clusters K							
Dataset	Hyperedge	Audited attributes	Class label	AHRC	AHCKA	GRAC	CAHC	CE+FL	Nodes	Hyperedges	Memberships
ML-1M	co-raters of one film	gender, age	dominant film genre	11	18	18	18	1,118	5,891	20,271	236,154
OA-DB	authors of one work	country, institution, career stage	primary-topic field	8	11	11	11	135	3,027	2,823	11,819
OA-CS-3+	authors of one work	country, institution, career stage	primary-topic field	34	26	26	26	2,478	24,286	8,219	44,525
Pokec	users sharing one hobby	sex, age, region	education level	34	6	6	34	5,733	70,144	21,486	170,189
IMDb-Cast	cast of one film	birth cohort, career stage	dominant film genre	485	23	23	23‡	6,658	80,000	154,159	918,945
ACS-PUMS-HH	one household	age, sex, race†, disability†	education level	26	9	9	9	22,706	59,452	20,539	59,452

these datasets, the data owner declares no minimum size, as in Section III-C, except in the analysis of declared sizes in RQ1. The households of ACS-PUMS-HH, which have 2 to 10 members, additionally retain size-2 hyperedges and are declared disjoint by their data owner. IMDb-Cast takes non-adult feature films from 1990 to 2025 with 3 to 50 actors or actresses among the IMDb principal credits, the 80,000 actors with the most such films among those with at least two, and the films that still have 3 to 50 of them. All data are public or de-identified by their providers, and on every dataset the node features $\mathbf{X}$ exclude the audited attributes and the class label. The audited attributes are protected, such as sex and race, or policy-relevant, such as country.

b) Cells.: A cell is one combination of method, dataset, and audited attribute, and an audit compares it with a budget. An attribute enters the 75 cells if at least 30% of memberships lie in hyperedges of two or more groups, so ACS-PUMS-HH excludes race and disability.

c) Methods.: We run four published AHC methods, AHRC [4], AHCKA [3], GRAC [58], and CAHC [5], with the artifact's implementations and settings, and CE+Fair-Louvain, our Louvain-style search on the clique expansion (CE), the weighted clique expansion with $\omega(k) = 1/\binom{k}{2}$, maximizing modularity minus a penalty on clusters whose attribute shares depart from the dataset's, in the spirit of modularity-based fairness [59]. On IMDb-Cast, CAHC exceeds memory in its first pretraining epoch, and we audit CAHC-init, the partition that k-means returns on the embeddings of its randomly initialized encoder. Its two cells pass uncertified, and excluding them changes no count of failing cells, misses, or certified cells at B. All releases come from the same filtered incidence, and each method returns its natural partition into the K clusters of Table III. We rerun the clustering of 57 cells with four further seeds.

d) Auditors.: We list auditors and budgets in Table IV. The reconstruction auditor takes the maximal cliques as hyperedges, the training-free baseline of hypergraph reconstruction [19], and the learned reconstruction is the supervised method of that work. The surrogates and the reconstruction auditor are pairwise auditors.

TABLE IV
AUDITORS AND BUDGETS OF THE EXPERIMENTS. AN AUDITOR PASSES A PARTITION WHEN ITS STATISTIC IS AT MOST THE BUDGET.

Auditor	Release	Budgets
Dyadic surrogate	unweighted clique	B, self-calibrated, 0.1, τ
Normalized-clique surrogate	normalized clique	B, self-calibrated, 0.1, τ
Reconstruction auditor	unweighted clique	B, self-calibrated, 0.1, τ
Learned reconstruction	unweighted clique, training hyperedges	B, 0.1
Certificate U_P	P, or T and degrees	B, scaled B, 0.1, τ
Certificates U_Q, $U_{Q,50}$	normalized clique	B, τ
CGP	node-preserving release	0.1, τ

TABLE V
GAP BETWEEN TRUE HGP AND THE DYADIC SURROGATE ON EACH DATASET, AND CELLS THAT FAIL THE BUDGETS B AND 0.1.

Dataset	Cells with $\mathrm{HGP}_{\mathrm{true}} > \mathrm{HGP}_{\mathrm{sur}}$	Median gap	Failures at B	Failures at 0.1
IMDb-Cast	10 of 10	0.061	0	10 of 10
ML-1M	8 of 10	0.043	6	9 of 10
Pokec	15 of 15	0.028	0	13 of 15
ACS-PUMS-HH	8 of 10	0.028	0	8 of 10
OA-DB	14 of 15	0.021	0	10 of 15
OA-CS-3+	13 of 15	0.007	0	0 of 15
All 75 cells	68 of 75	0.024	6	50 of 75

e) Budgets.: Budgets are inputs of the audit, fixed for each dataset–attribute pair and shared across methods, as a declared limit applies to every partition regardless of its number of clusters. The shared budget B follows the random-allocation benchmark of segregation measurement in small units, which compares observed segregation with that of a random allocation of individuals to hyperedges [60]. It is the 95th percentile of HGP over 200 partitions that preserve the cluster sizes of the AHRC partition and reassign nodes at random within strata, which are the class labels of Table III and not the audited attribute. We also compute the 95% bootstrap interval of this percentile. Computing B needs the incidence and the

attributes in one place, so it only defines the true verdicts. A self-calibrated budget is the 95th percentile of the statistic of the auditor over the same reassignments, and a partition-calibrated budget uses the cluster sizes of the audited partition. We scale the shared and the self-calibrated budgets by nine factors from 0.6 to 2.0 and take the true verdict at the same scaled budget.

f) Declared budgets.: A deployment instead declares a budget, which needs no joint access to the incidence and the attributes. On the dissimilarity scale, an auditor may declare any budget τ on $\bar{D}$ or on $D_{\mathrm{CGP}} = \mathrm{CGP}/(1 - \sum_g p_g^2)$, the generalized dissimilarity index with the clusters as areal units [32]. We sweep τ from 0.1 to 0.7, which contains the reference values 0.3 and 0.6 of Massey and Denton [61, p. 20]. Random allocation gives $\bar{D}$ a 95th percentile of 0.56 to 0.90 on the 15 dataset–attribute pairs, as small hyperedges are uneven by chance [60], so we compare each auditor with the true verdict at every declared τ. The fixed budget is 0.1 on HGP and CGP, the bound within which practice [62], [63] treats the statistical parity difference of toolkits such as AI Fairness 360 [64] as fair.All budgets are reference points of a sweep, not normative limits.

g) Metrics.: A cell passes when $\mathrm{HGP}_{\mathrm{true}} \le B$ and fails otherwise. A *miss* is a failing cell that an auditor certifies, and a *false alarm* is a passing cell that it fails. A conservative certificate U returns *certified* when $U \le B$ and *undetermined* otherwise, and undetermined cells go to exact evaluation. The *coverage* of a certificate is the fraction of passing cells that it certifies. Under a budget τ, a cell fails when $\bar{D} > \tau$, each pairwise auditor divides its HGP by the $\bar{I}$ of the hyperedges that it assumes, and the certificates are those of Proposition 3.

h) Reproducibility and hardware.: With the raw data, the artifact regenerates every release, budget, audit, certificate, and count of RQ1 and RQ2 and verifies every witness. The 75 cells run on an Intel Core i9-14900KF workstation with 32 GB RAM. The artifact stores the 30 partition files, the processed OpenAlex hypergraphs, and the search witnesses, and it evaluates every decision in exact rational arithmetic, with outward rounding on floating-point releases. Trained CAHC on Pokec and ACS-PUMS-HH runs on a server with 22 CPUs and 110 GiB, and the prototype on SQLite 3.50.4 and DuckDB 2.0.0.

B. RQ1: Information Lost by Pairwise Releases

To answer RQ1, we search for inputs that share a release but not the verdict, first for cluster-level counts and then for pairwise graphs, and compare pairwise auditors with the true verdicts on the same partitions.

- **Same-group exchanges turn every passing cell into a failing one.** A hill-climbing search over such exchanges, which preserve every audit of cluster-level counts by Remark 1, makes every passing cell fail after reassigning a median of 10.4% of the nodes. It raises HGP by a median of 0.22 over all cells.
- **The unweighted clique leaves many verdicts undetermined even when the number and sizes of hyperedges are declared.** With hyperedge sizes alone declared, replacing hyperedges of $k \ge 4$ members by their $\binom{k}{3}$ triples preserves the unweighted clique and leaves the verdict undetermined in 41 of the 65 cells with overlapping hyperedges. When the number of hyperedges and the size of every hyperedge are also declared, member swaps that preserve these quantities and the clique find compatible inputs on both sides of the budget in at least 24 of these cells, including all six failing cells, even when every degree is declared, as shown in Figure 3. In every undetermined cell, each release-only auditor errs on some compatible input.
- **The dyadic surrogate understates true HGP** in 68 of the 75 cells, on all six datasets, with a median gap of 0.024 and with the per-dataset values in the extended version [44], in the direction of the gap in Theorem 3. In some of the 65 cells with overlapping hyperedges, the worst invitation policy found creates an average gap above $\mathrm{HGP}_{\mathrm{sur}}$, but HGP bounds every policy by Proposition 1. On ACS-PUMS-HH, whose households are declared disjoint, the connected components of the unweighted clique give HGP exactly in all 10 cells.
- **Pairwise auditors misjudge cells at every budget.** At B,

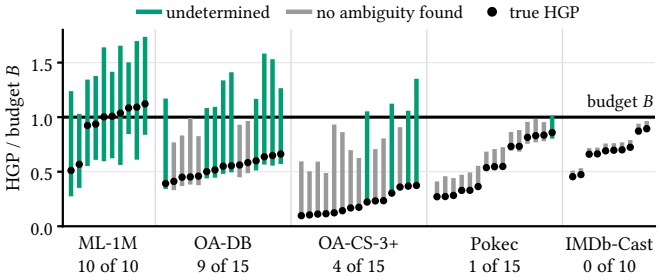


Fig. 3. Range of HGP over the compatible inputs in each cell with overlapping hyperedges, with their number and sizes declared. Dots mark true HGP.

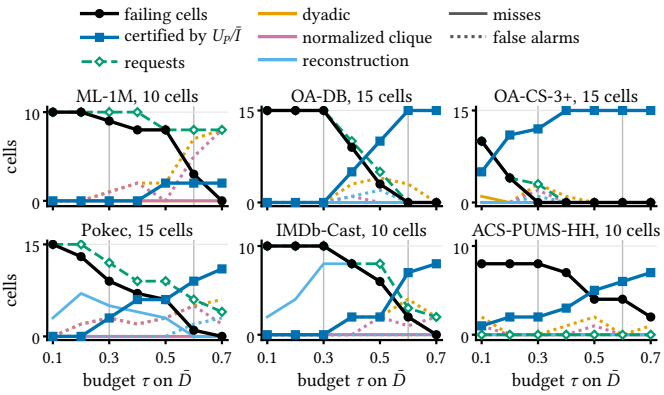


Fig. 4. Verdicts per dataset as the budget τ on $\bar{D}$ grows, with the reference values 0.3 and 0.6 in gray. Solid lines count misses, dotted lines false alarms, and requests the cells sent for exact evaluation. Reconstruction is not run on ML-1M.

TABLE VI

VERDICTS ON THE 75 CELLS, ONE BUDGET PER ROW. FAIL COUNTS THE CELLS ABOVE THE BUDGET, A MISS IS A FAILING CELL THAT AN AUDITOR PASSES, AND A FALSE ALARM IS A PASSING CELL THAT IT FAILS. DY., NC, AND REC. ARE THE DYADIC SURROGATE, THE NORMALIZED-CLIQUE SURROGATE, AND THE RECONSTRUCTION AUDITOR, NOT RUN ON ML-1M.

		Misses ↓				False alarms ↓				
Budget	Fail	Dy.	NC	Rec.	U_P	Dy.	NC	Rec.	U_P certifies ↑	Requests
Shared B	6	6	6	–	0	0	0	3	62 of 69	13
Self-calibrated	6	1	5	–	–	0	0	3	–	–
Fixed 0.1	50	6	9	9	0	1	0	2	23 of 25	44
$\tau = 0.3$ on $\bar{D}$	51	0	0	13	0	7	6	1	17 of 24	50
$\tau = 0.6$ on $\bar{D}$	10	0	0	2	0	19	11	2	54 of 65	17

both surrogates miss all six failing cells, all on ML-1M and beyond the bootstrap interval of the budget, and the misses persist in every further seed. The self-calibrated surrogates still miss failing cells. With B scaled by 0.6 and at the fixed budget 0.1, the misses extend to OA-DB, Pokec, and IMDb-Cast. On the dissimilarity scale, the reconstruction auditor misses failing cells at each τ up to 0.5, and both surrogates fail passing cells at each τ, as counted in Table VI.

- **With maximal cliques as hyperedges, the reconstruction auditor misestimates HGP** by -0.23 to $+0.27$ on Pokec and IMDb-Cast and fails passing cells of CE+Fair-Louvain on Pokec under the shared and self-calibrated budgets. The learned reconstruction of Wang and Kleinberg [19], run with the authors' code, matches every true verdict only when trained on half the hyperedges of the audited hypergraph, which a release-only auditor lacks. Trained on another OpenAlex hypergraph instead, it gives the same two false alarms as the reconstruction auditor on OA-DB at 0.1. With half the audited hyperedges, it matches at the shared budget and at 0.1.
- **Shares of memberships in split hyperedges do not fix HGP.** On ML-1M, the partitions of GRAC and CE+Fair-Louvain differ widely in number of clusters but place nearly the same share of memberships, 0.916 and 0.923, in split hyperedges. Their HGP differs about twofold for both attributes. Refining a partition never lowers HGP by Remark 2, but no natural partition of a dataset refines another.

C. RQ2: Certification from a Held Release

To answer RQ2, we certify the budgets of the same cells from the random-walk release and the normalized clique, count the audits that still need exact counts, and measure the scale of exact evaluation and the size of the requests.

- **The random-walk release certifies most passing cells.** At B and on the dissimilarity scale, U_P and $U_P/\bar{I}$ do so with no declaration and no cap on hyperedge size, as listed in Table VI. At B, U_P certifies 62 of the 69 passing cells, more than $U_{Q,50}$ and U_Q, and still 52 of the 59 outside ACS-PUMS-HH, whose release determines HGP. This coverage holds across datasets and methods, with every passing cell certified on OA-DB, OA-CS-3+, and ACS-PUMS-HH, at least half on each other dataset, and at least 11 of the 13 to 15 passing cells of each method. On the dissimilarity scale, the normalized clique certifies fewer than $U_P/\bar{I}$ even under the declared sizes, as shown in Figure 4. No certificate of Propositions 2 and 3 certifies a failing cell at any budget. U_P certifies a similar share under partition-calibrated budgets and in every further seed.
- **Exact counts are requested only for the remaining audits.** The households of ACS-PUMS-HH are declared disjoint, so the release determines HGP on its 10 cells, and line 1 of Algorithm 1 decides them at every budget. The other audits need a request when they fail or when U_P leaves them uncertified, as counted in Table VI. Certify-then-request avoids exact evaluation in 62 of the 75 audits at B, but in fewer under the budget 0.1 and $\tau = 0.3$, as most cells fail under them. At B and at $\tau = 0.6$, 9 and 8 of the 15 dataset–attribute pairs need no request at all. The passing cells left uncertified lie close to the budget, with HGP 6% to 19% below B against a median margin of 53% for the certified ones, and deciding $h_{\max} > B$ is NP-complete in the worst case by Theorem 4.
- **The bound U_P remains close to true HGP**, with a median ratio to HGP of 1.12 against 1.38 for $U_{Q,50}$. By Lemma 1, U_P exceeds HGP only through hyperedges of four or more members in which neither clusters nor groups are a function of the other, and such hyperedges hold at least half of the memberships in the 7 uncertified passing cells, against a median of 12% in the certified ones. On 1,500 subhypergraphs of 8 to 40 nodes grown from the 75 cells, the tightest certificate $h_{\max}$ equals HGP, and U_P exceeds it by a median factor of 1.04. The relaxation of Proposition 4 equals $h_{\max}$ wherever it finishes but exceeds our cap of 300,000 candidate hyperedges when large hyperedges appear, while U_P needs one pass over the release.
- **The certificates retain their coverage on rounded and pruned releases.** Under Proposition 5, rounded releases certify as many cells as exact ones, and U_P certifies 55 when each row of T retains its 50 largest entries, where the share of crossing mass among the retained entries alone would certify two failing ML-1M cells, a false certificate that part (b) prevents.

a) Scale-up.: We use a larger IMDb-Cast hypergraph, which has 5.70M memberships against 0.92M in Table III, together with its disjoint copies, which preserve its HGP. The hypergraph covers films and TV series with three or more principal credits, and the copies preserve its degree and hyperedge-size distributions. With 6 threads and a DuckDB memory limit of 4 GB, our exact SQL query returns HGP in 69.8 s for $2.05 \cdot 10^8$ memberships at a peak memory of 4.2 GiB, and its time grows with a log–log slope of 1.14 up to $4.6 \cdot 10^7$ memberships. At the largest size, the exact random-walk release takes 161 s to build and occupies 12.5 GB, and its U_P scan takes 58.8 s. On a single copy with the AHRC partition, the release takes 1.2 s to build and occupies 347 MB, and U_P certifies both audited attributes at budgets computed

as B.

b) Incremental maintenance.: Maintaining the per-hyperedge counts updates exact HGP in a median of at most $240\,\mu$s per single-node move and U_P in at most $64\,\mu$s, against at least 0.06 s for full recomputation.

c) Batch audits.: In batch audits of the 75 cells with nine scaled budgets, the gain of certification is that a certified audit needs no joint processing of attributes and incidence. One request fixes the verdict at every budget, and a data owner that may retain the attribute column needs only the partition column for later partitions. Likewise, one joint computation per dataset–attribute pair fixes B for every partition, and a declared τ needs none. Over 10,800 audits of natural and perturbed partitions, certify-then-request issues no false certificate or wrong verdict. Where the pipeline lacks the release, the data owner builds one of at most 36 MB within 0.25 s, and each dataset's requests move at most 2 MB out and 8 KB back. Certify-then-request with the release build and exact evaluation of every audit both finish a dataset within 1.5 s.

VI. Related Work

View determinacy. View determinacy and certain answers [11], [12], [28], [65]–[70], studied at PODS and VLDB among others, concern the queries that relational views answer, and our release is the view and HGP the aggregate query. Rewriting aggregate queries using views [66], [71] is the closest database setting. On multihypergraphs, our criterion reduces to the span condition of SUM-query auditing [37], [38]. Auditing sum queries for the disclosure of individual Boolean values is computationally hard [72]. Our certificates are dual solutions of a linear relaxation, like fractional edge-cover bounds on join sizes [73] and contingency-table bounds [74], [75]. Beyond these settings, we study a fairness aggregate over pairwise releases of hypergraphs, locate its determinacy boundary between three and four members, and show that its tightest certificate is NP-hard.

Responsible data management. Fairness work at VLDB, SIGMOD, and ICDE [76] assesses dataset coverage [77], repairs training data [78], surveys fair rankings and recommendations [79], and builds fair hashmaps [80] and fair max-min diversification [81], with the data at the analyzing party, but our auditor holds only a release.

AHC. Prior work on AHC and hypergraph data management, including work at VLDB and SIGMOD, optimizes partitions or systems [1]–[5], [58], [82], [83] and defines no fairness audit, so our audits apply to the partitions and releases that it computes. Reconstruction from projections [19], [84] infers one likely hypergraph, and recovery thresholds for random hypergraphs [85] concern all hyperedges, while determinacy concerns one aggregate and a certificate bounds it on every compatible input.

Fair clustering and auditing. Outside the database work above, fair clustering constrains clusters, as in fairlets [24], fair spectral clustering [86], federated fair k-means [26], clustering on heterogeneous information networks [25], proportional fairness of centers [87], [88], and modularity-based fairness in communities [59], [89], all on clusters or centers, but HGP measures fairness inside hyperedges. Auditing work tests fairness by betting [27], audits conditional fairness by optimal transport [90] and subgroup fairness [91], and certifies disparate impact [92], all on observable audit values. To our knowledge, no prior work certifies a fairness audit of clustering from a release without hyperedges. Auditing with unobserved protected attributes [93] bounds attribute uncertainty, not hyperedge identity.

Governance, privacy, and verification. Governance and privacy work [94]–[99] specifies the data that a data owner may release. Certifying algorithms and verifiable SQL [100]–[104] verify computations on given data, so on a pairwise graph they confirm only the dyadic surrogate and assume determinacy.

VII. Conclusion

We study the conditions under which releases determine HGP in AHC. Up to three members per hyperedge, the random-walk release determines HGP, and so do exact weighted clique expansions at a declared uniform size. For larger hyperedges, the random-walk release still gives a conservative certificate. A certified budget limits the average within-hyperedge gap of Proposition 1 to at most that budget. Computing the tightest certificate is NP-hard, and per-size counts resolve the remaining budgets exactly. On six real-world datasets, pairwise auditors misjudge partitions in both directions wherever hyperedges overlap. The random-walk certificate certifies most passing partitions without access to the hyperedges, and exact evaluation scales to hundreds of millions of memberships.

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APPENDIX

This appendix gives the complete proofs of the results whose proofs the main text shortens or sketches, and further verdicts in Appendix F0a. Numbers of theorems, lemmas, propositions, corollaries, equations, figures, and tables refer to the main text.

A. Proof of Lemma 1

Proof. Let $k = |e|$ be the number of members of e. For a group g and a cluster C, let $n_{11} = n_{e,g,C}$, and let n_{10}, n_{01}, and n_{00} count the members of e that are in g but outside C, in C but outside g, and in neither, with $k = n_{11}+n_{10}+n_{01}+n_{00}$. By Eq. (3), $\psi_e = \frac{1}{2k^2}\sum_{g,C}|R_{gC}|$ with the residual $R_{gC} = kn_{11} - (n_{11}+n_{10})(n_{11}+n_{01}) = n_{11}n_{00} - n_{10}n_{01}$. As $n_{11}+n_{10}$ depends only on g and $n_{11}+n_{01}$ only on C, we have $\sum_{g,C}(n_{11}+n_{10})(n_{11}+n_{01}) = k\cdot k = \sum_{g,C} kn_{11}$. The values R_{gC} sum to 0, and $R_{gC} = -n_{10}n_{01} \le 0$ whenever $n_{11} = 0$. It follows that $\sum_{g,C}|R_{gC}| = 2\sum_{g,C}\max(R_{gC}, 0) \le 2\sum_{g,C} n_{11}n_{00}$, given that $n_{10}n_{01} \ge 0$, and each term is tight when $n_{11}n_{00} = 0$ or $n_{10}n_{01} = 0$. A member in group g and cluster C differs in both group and cluster from exactly n_{00} members, so $\sum_{g,C} n_{11}n_{00}$ counts each such pair twice and equals $2N_e$. This proves $\psi_e \le 2N_e/k^2$. If the clusters inside e are a function of the groups, then $n_{10} = 0$ whenever $n_{11} > 0$, and if the groups are a function of the clusters, then $n_{01} = 0$ whenever $n_{11} > 0$, so equality holds in both cases. A cell with $n_{11}n_{00} > 0$ and $n_{10}n_{01} > 0$ needs four members, so equality also holds if $k \le 3$. □

B. Proof of Corollary 3

Proof. (i) The equations of the hyperedges of two members force $\eta_{uv} = \chi_{uv}/\omega(2)$ for $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$ and $\eta_{uv} = 2/\omega(2)$ for $\mathrm{vol}(\mathcal{H})$, and the equation of f then reads $(p/q)N_f = 2N_f/3$ and $6p/q = 3$, respectively, which are the exact conditions by Theorem 1(a) and (b). Both families give each pair of f the weight $p\,\omega(2)$. The HGP of q copies of f is $\psi_f = 2N_f/9$, and that of p copies of each pair of f is $N_f/6$, as the three pairs have equal weight and each crossing pair has $\psi = 1/2$.

(ii) The hyperedge f and its three pairs have the same unweighted clique and the HGP values $2N_f/9$ and $N_f/6$ of (i). On four or more nodes, take a crossing pair ut, which exists given that f contains one, and two further nodes v and w. The inputs $\{uvw, ut, vt, wt\}$ and $\{uvt, uwt, vwt\}$ have the same normalized clique, as each pair inside uvw receives $1/2$ and each pair at t receives 1 from either side. Their values of $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP}$ differ by one third of the number of crossing pairs between t and uvw, which is at least one, and the normalized clique fixes $\mathrm{vol}(\mathcal{H})$. The two inputs above are the case $uvw = 012$ and $t = 3$. □

C. Proof of Proposition 3(b)

Proof. (b) A hyperedge e adds $N_e/(|e|-1)$ to $\sum_{uv} w_{uv}\chi_{uv}$ and $|e|^2 I_e/(2|e|-2)$ to $\sum_{uv} w_{uv}\xi_{uv}$, whereas $\mathrm{vol}(\mathcal{H})\,\mathrm{HGP} \le \sum_e 2N_e/|e|$ by Lemma 1 and $\mathrm{vol}(\mathcal{H})\,\bar{I} = \sum_e |e|\,I_e$. As $(k-1)/k$ increases in k, $\sum_e 2N_e/|e| \le 2\frac{\bar{r}-1}{\bar{r}}\sum_{uv} w_{uv}\chi_{uv}$ and $\sum_e |e|\,I_e \ge 2\frac{r_{\min}-1}{r_{\min}}\sum_{uv} w_{uv}\xi_{uv}$, and dividing gives (b). For the constants, $U_P = \mathrm{HGP}$ on H_1, and one hyperedge of $\bar{r}$ members whose groups are its clusters, together with k hyperedges of $r_{\min}$ members that lie in one cluster and hold two groups, gives a ratio of the two sides that tends to c as k grows, and c tends to 2 as $\bar{r}$ grows with $r_{\min} = 2$. □

D. Proof of Theorem 3

With $s = \lceil r/2 \rceil$ and $t = \lfloor r/2 \rfloor$, the gap of Theorem 3 is $\gamma(r) = st(r-2)/(r^2(r-1))$. For even r, $\gamma(r) = (r-2)/(4(r-1))$, and for odd r, $\gamma(r) = (r+1)(r-2)/(4r^2)$. Any estimator from the clique returns one value on both inputs, so one error is at least $\gamma(r)/2$.

Derivation. Split the r nodes into blocks of $s = \lceil r/2 \rceil$ and $t = \lfloor r/2 \rfloor$ nodes, where the first forms group a and cluster C_1 and the second forms group b and cluster C_2. The first hypergraph has one hyperedge with all r nodes, and the second has the $\binom{r}{2}$ pairs as hyperedges, and both have the unweighted clique K_r.

Single hyperedge. The hyperedge has $q_e = (s/r, t/r)$, $q_{e,a} = (1, 0)$, and $q_{e,b} = (0, 1)$, which gives $\mathrm{TV}(q_{e,a}, q_e) = t/r$ and $\mathrm{TV}(q_{e,b}, q_e) = s/r$. With $p_{a|e} = s/r$ and $p_{b|e} = t/r$ this gives $\psi_e = \frac{s}{r}\cdot\frac{t}{r} + \frac{t}{r}\cdot\frac{s}{r} = \frac{2st}{r^2}$, and since there is one hyperedge, $\mathrm{HGP} = 2st/r^2$.

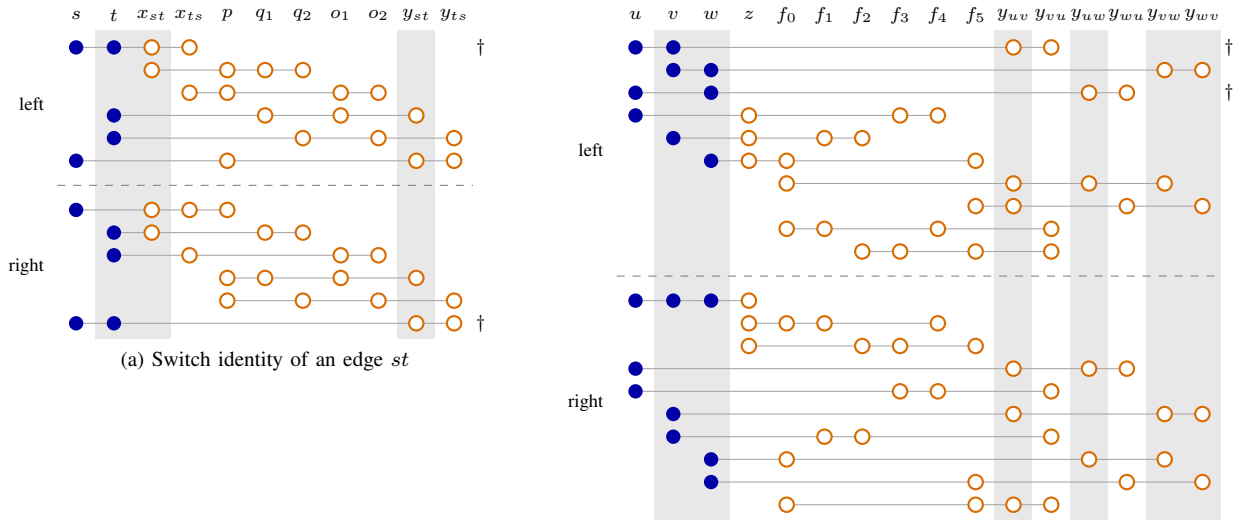
All pairs. A same-group pair lies in one cluster, so each of its groups is spread like the hyperedge and $\psi = 0$. A cross-group pair has its members in different clusters, so $q_e = (1/2, 1/2)$, each group is a point mass, both total-variation terms equal $1/2$, and $\psi = 1/2$. Each pair has weight $2/(r(r-1))$ and there are st cross-group pairs, giving $\mathrm{HGP} = st\cdot\frac{1}{2}\cdot\frac{2}{r(r-1)} = \frac{st}{r(r-1)}$.

Difference. The two values differ by $\frac{2st}{r^2} - \frac{st}{r(r-1)} = \frac{st\,[2(r-1)-r]}{r^2(r-1)} = \gamma(r)$. Evaluating at $r = 3, 4, 32$ gives the stated values, and since $st/r^2 \to 1/4$ and $(r-2)/(r-1) \to 1$, the difference tends to $1/4$.

E. Proof of Theorem 4

Proof. Let μ_{uv} count the hyperedges that contain u and v, and d_v the hyperedges that contain v. The release fixes μ in both parts. A compatible input has hyperedges of at most R members, where $R = r$ in (a), so it has at most $\sum_{k\le R}\binom{n}{k}$ distinct hyperedges, each with a multiplicity at most μ_{uv} for every pair uv in it. The set of compatible inputs is finite and $h_{\max}$ is attained. One compatible input, listed by its distinct hyperedges and multiplicities, is a certificate of polynomial size whose release and HGP are checked exactly, which gives membership in NP, also for $h_{\max} = U_P$, given that $\mathrm{HGP} \le U_P$ by Proposition 2(a).

Part (a). We reduce from completing a partial Latin square with $\ell \ge 1$ empty cells [50], as a complete square maps to a fixed yes-instance. Its defect graph $G = (V_G, E_G)$ has the parts V_1, V_2, V_3 of rows, columns, and symbols, joins row i to column j if cell ij is empty, and joins each row or column to the symbols that it lacks, so it has ℓ edges between any two parts. A triangle has one node in each part and fills one cell, and the square can be completed exactly when the edges of G partition into triangles. The reduction gives every edge of G that meets V_1 the same excess of U_P over HGP, and a hyperedge that covers a triangle removes the excess of two



(a) Switch identity of an edge st

(b) Absorber identity of a triangle uvw

Fig. 5. Gadgets of the reduction in the proof of Theorem 4(a) at $r = 4$. Columns are nodes, filled for group a and hollow for group b, with cluster C_2 shaded, and rows are hyperedges. The rows above the dashed line form the left side of an identity and the rows below it the right side, and both sides cover the same pairs, each once. (a) The switch of an edge st with $s \in V_1$ and $t \in V_2$, whose left side lies in $\mathcal{H}_0$. (b) The absorber of a triangle uvw with $u \in V_1$, whose left side consists of the hyperedges Y_{uv}, Y_{vw}, Y_{uw} that the switches produce and of seven hyperedges of $\mathcal{H}_0$. A dagger marks a hyperedge whose slack $U_P -$ HGP contributes $1/4$, and every other hyperedge contributes 0.

such edges, so $h_{\max}$ reaches U_P exactly when the triangles chosen partition the edges. We order V_G. For each edge st with $s < t$, we add nodes x_{st}, x_{ts} and the hyperedge $s\,t\,x_{st}\,x_{ts}$, and for each triangle uvw with $u < v < w$, private nodes $z, f_0, \ldots, f_5$ and, for each of its edges st with $s < t$, private nodes $p, q_1, q_2, o_1, o_2, y_{st}, y_{ts}$. We write $Y_{st} = s\,t\,y_{st}\,y_{ts}$ and $\mathcal{A} \equiv \mathcal{B}$ if two families of hyperedges cover the same pairs, each once. Every edge st of a triangle uvw satisfies the switch identity in the first four lines below, and the triangle satisfies the absorber identity in the rest. Both identities are drawn in Figure 5.

$$
\begin{aligned}
&s\,t\,x_{st}\,x_{ts},\ x_{st}\,p\,q_1\,q_2,\ x_{ts}\,p\,o_1\,o_2,\\
&\quad t\,q_1\,o_1\,y_{st},\ t\,q_2\,o_2\,y_{ts},\ s\,p\,y_{st}\,y_{ts}\\
&\quad \equiv s\,x_{st}\,x_{ts}\,p,\ t\,x_{st}\,q_1\,q_2,\ t\,x_{ts}\,o_1\,o_2,\\
&\quad p\,q_1\,o_1\,y_{st},\ p\,q_2\,o_2\,y_{ts},\ Y_{st},\\
&Y_{uv},\ Y_{vw},\ Y_{uw},\ u\,z\,f_3\,f_4,\ v\,z\,f_1\,f_2,\ w\,z\,f_0\,f_5,\\
&\quad y_{uv}\,y_{uw}\,y_{vw}\,f_0,\ y_{uv}\,y_{wu}\,y_{wv}\,f_5,\ y_{vu}\,f_0\,f_1\,f_4,\\
&\quad y_{vu}\,f_2\,f_3\,f_5 \equiv u\,v\,w\,z,\ z\,f_0\,f_1\,f_4,\ z\,f_2\,f_3\,f_5,\\
&\quad u\,y_{uv}\,y_{uw}\,y_{wu},\ u\,y_{vu}\,f_3\,f_4,\ v\,y_{uv}\,y_{vw}\,y_{wv},\\
&\quad v\,y_{vu}\,f_1\,f_2,\ w\,y_{uw}\,y_{vw}\,f_0,\ w\,y_{wu}\,y_{wv}\,f_5,\\
&\quad y_{uv}\,y_{vu}\,f_0\,f_5.
\end{aligned}
$$

The given compatible input is the witness $\mathcal{H}_0$, which holds every hyperedge $s\,t\,x_{st}\,x_{ts}$, the other five left hyperedges of every switch identity, and the seven left hyperedges other than Y_{uv}, Y_{vw}, Y_{uw} of every absorber identity, and m is its number of hyperedges. The graph G has at most $n\ell$ triangles for a square of order n, so the construction takes polynomial time. Group a is V_G and group b holds the new nodes. A node x_{st} or y_{st}, in either order of s and t, takes the cluster of its second index, which places x_{st} and y_{st} in the cluster of t and x_{ts} and y_{ts} in the cluster of s. Cluster C_2 holds $V_2 \cup V_3$ and every node x_{st} or y_{st} whose second index lies in $V_2 \cup V_3$, and C_1 holds every other node. We release the weighted clique or P of $\mathcal{H}_0$ and set $B = U_P - 1/(4m)$.

Every hyperedge of $\mathcal{H}_0$ other than $s\,t\,x_{st}\,x_{ts}$ has at most one member in V_G or among the nodes x_{st}, its other members are private to one triangle, and the hyperedges built for one triangle pairwise share at most one node. This gives $\mu_{uv} \le 1$. As $w_{uv} = \omega(4)\mu_{uv}$, $P_{uv} = \mu_{uv}/4$, and $P_{vv} = \sum_u \mu_{uv}/12$, either release fixes μ, and the compatible inputs are the partitions of $\Gamma = \{uv : \mu_{uv} = 1\}$ into 4-cliques. Here Γ restricted to V_G is G. The common neighbors of an edge st outside V_G are x_{st}, x_{ts} and the nodes y_{st}, y_{ts}, z of each triangle through st, and among them the adjacent pairs are $x_{st}x_{ts}$ and the pairs $y_{st}y_{ts}$ of one triangle. Among these common neighbors, a node adjacent to a third node $w \in V_G$ is the node z of the triangle stw. A 4-clique with two nodes of V_G is $s\,t\,x_{st}\,x_{ts}$ or some Y_{st}, one with three is $uvwz$, and none has four, as G is tripartite. All 4-cliques of Γ are listed in Lemma 2 after this proof.

For a hyperedge with α_1, α_2 members of group a and β_1, β_2 of group b in C_1, C_2, every entry of Eq. (3) is $\pm(\alpha_1\beta_2 - \alpha_2\beta_1)/r^2$ and $N_e = \alpha_1\beta_2 + \alpha_2\beta_1$. With $U_P = \frac{1}{m}\sum_e 2N_e/r^2$ from the proof of Proposition 2, we obtain $U_P -$ HGP $= \frac{1}{m}\sum_e \sigma_e$ with $\sigma_e = 4\min(\alpha_1\beta_2, \alpha_2\beta_1)/r^2$, which vanishes if one group of the hyperedge lies in one cluster. A 4-clique with at most one node of V_G has at most one member of group a, and the hyperedges $uvwz$, $s\,t\,x_{st}\,x_{ts}$, and Y_{st} with $s, t \in V_2 \cup V_3$ have one group inside one cluster, so all of

these have $\sigma_e = 0$. The 4-cliques with $\sigma_e > 0$ are exactly the hyperedges $s\,t\,x_{st}\,x_{ts}$ and Y_{st} whose edge st is one of the 2ℓ edges incident to V_1, and they have $\alpha_1 = \alpha_2 = \beta_1 = \beta_2 = 1$ and $\sigma_e = 1/4$. A hyperedge $uvwz$ covers two such edges, and the triangles of the λ hyperedges $uvwz$ of a compatible input are edge-disjoint, so $\mathrm{HGP} = U_P - (\ell - \lambda)/(2m)$. Conversely, for edge-disjoint triangles, replacing the left sides of their switch and then absorber identities in $\mathcal{H}_0$ by the right sides uses each hyperedge $s\,t\,x_{st}\,x_{ts}$ at most once and preserves μ. This gives $h_{\max} = U_P - (\ell - \nu)/(2m)$ for the largest number ν of edge-disjoint triangles, which is $U_P > B$ if the square can be completed and at most $U_P - 1/(2m) < B$ otherwise.

For $r \geq 5$, we add the same set Z of $r-4$ nodes of group b in C_1 to every hyperedge of $\mathcal{H}_0$ and set $B = U_P - 4/(r^2 m)$. The release fixes m through $\sum_{u<v} \mu_{uv} = m\binom{r}{2}$, and for $z' \in Z$ the multiplicities $\mu_{z'v}$ sum to $(r-1)m$, which is $r-1$ times the degree of z' in a compatible input, so every hyperedge contains Z. Removing Z gives a 4-clique partition of the graph Γ of the unpadded $\mathcal{H}_0$, and padding one reproduces μ, as $\mu_{z'v} = d_v = \sum_{u \notin Z} \mu_{uv}/3$. Padded crossing hyperedges have $(\alpha_1, \alpha_2, \beta_1, \beta_2) = (1, 1, r-3, 1)$ and $\sigma_e = 4/r^2$, and the other hyperedges keep $\sigma_e = 0$, so $\mathrm{HGP} = U_P - 8(\ell - \lambda)/(r^2 m)$ and B separates the two cases. For every $r \geq 4$, a bound U with $h_{\max} \leq U < h_{\max} + 8/(r^2 m)$ satisfies $U \geq U_P$ exactly when the square can be completed.

Part (b). We reduce from partitioning the edges of a graph $G = (V_G, E_G)$ into copies of K_R [49], with $M' = |E_G| \geq 1$, as an edgeless graph maps to a fixed yes-instance. We place V_G in group a and C_1, add M' disjoint pairs $x_i y_i$ with x_i in group a and C_1 and y_i in group b and C_2, and release $\omega(2)$ on every edge of G and pair $x_i y_i$. As $\omega(k) = \omega(2)$ for $k \leq R$, every released pair lies in exactly one hyperedge and no other pair lies in a hyperedge. The compatible inputs consist of the pairs $x_i y_i$ and of cliques of two to R nodes of G that partition E_G, and the input with one size-2 hyperedge per pair is among them. Hyperedges inside V_G have $\psi_e = 0$ and each pair $x_i y_i$ has $\psi_e = 1/2$, so $\mathrm{HGP} = M'/(\mathrm{vol}_G + 2M')$ with $\mathrm{vol}_G = \frac{2M'}{R-1} + \sum_e \frac{|e|(R-|e|)}{R-1}$ over the cliques, where a term vanishes if $|e| = R$ and is at least one otherwise, as $(|e|-1)(R-|e|-1) \geq 0$. With $S^* = 2M'R/(R-1)$, which satisfies $S^* \leq 3M'$ and $S^* + 1 \leq 4M'$, we have $h_{\max} = M'/S^*$ if G has a K_R-partition and $h_{\max} \leq M'/(S^*+1)$ otherwise. The budget $B = 2M'/(2S^*+1)$ separates the cases, and the gap $M'/(S^*(S^*+1))$ is at least $1/(12M') = 1/(6M)$ for the $M = 2M'$ released pairs, which gives the bound on U as in (a). □

We next add the details that this proof states without derivation.

a) 4-cliques of Γ.: The proof needs only the 4-cliques with two or more nodes of V_G and the fact that every other 4-clique has at most one member of group a, and the next lemma lists all 4-cliques.

Lemma 2 (4-cliques of the pair graph)**.** *In the reduction of part (a) at $r = 4$, the 4-cliques of $\Gamma = \{uv : \mu_{uv} = 1\}$ are exactly the hyperedges that occur on either side of a switch identity or an absorber identity. A 4-clique with two nodes of V_G is a hyperedge $s\,t\,x_{st}\,x_{ts}$ or a hyperedge Y_{st} of a triangle through st, a 4-clique with three nodes of V_G is a hyperedge $uvwz$, no 4-clique has four, and every other 4-clique has at most one node of V_G.*

Proof. Call the nodes of V_G and the nodes x_{st} shared, and every other node private to the triangle for which it is added. Every hyperedge of $\mathcal{H}_0$ other than $s\,t\,x_{st}\,x_{ts}$ has at most one shared member, and its other members are private to one triangle. Hence two shared nodes are adjacent in Γ only inside a hyperedge $s\,t\,x_{st}\,x_{ts}$, two private nodes only if they are private to the same triangle, and a node private to a triangle uvw only to private nodes of uvw and to the shared nodes u, v, w and x of the three edges of uvw.

Let Q be a 4-clique of Γ. If all nodes of Q are shared, its pairs lie in hyperedges $s\,t\,x_{st}\,x_{ts}$. A node x_{st} lies in one such hyperedge, so Q is that hyperedge if it contains a node x, and otherwise Q is a 4-clique of G, which does not exist as G is tripartite. If Q contains a node private to a triangle T, all nodes of Q are private to T or shared nodes of T, so Q lies in the closed gadget of T, which consists of the three nodes of T, the six nodes x of its edges, and its 28 private nodes. On this set, Γ consists of the pairs of the three hyperedges $s\,t\,x_{st}\,x_{ts}$ of the edges of T and of the 22 hyperedges of $\mathcal{H}_0$ built for T, as a hyperedge built for another triangle meets this set in at most one shared node. Up to the names of its nodes, this graph is the pair graph of the reduction for the partial Latin square of order one with its cell empty, whose witness consists of exactly these 25 hyperedges. Exhaustive enumeration on this graph with 37 nodes and 150 pairs finds 53 4-cliques, which are the three hyperedges $s\,t\,x_{st}\,x_{ts}$, the five other left hyperedges and the six right hyperedges of each switch identity, among them the hyperedges Y_{st}, and the seven left hyperedges of $\mathcal{H}_0$ and the ten right hyperedges of the absorber identity. Conversely, each hyperedge of an identity is a 4-clique of Γ, as the two sides of an identity cover the same pairs and the left sides cover pairs of Γ. The counts of nodes of V_G in these hyperedges give the second statement. □

b) Smallest instances.: For the partial Latin square of order one with its cell empty, G is one triangle, so $\ell = 1$, and $\mathcal{H}_0$ has $m = 25$ hyperedges on 37 nodes. Exhaustive enumeration in exact arithmetic finds 9 partitions of Γ into 4-cliques, whose HGP values are $\mathrm{HGP}(\mathcal{H}_0) = 23/200$ and $U_P = 27/200$, so $h_{\max} = U_P$ and $U_P - \mathrm{HGP}(\mathcal{H}_0) = \ell/(2m) = 1/50$. For the square of order two with one filled cell, $\ell = 3$ and $m = 97$, and the 2,261 partitions take exactly the four values $U_P - (3-\lambda)/(2m)$ for $\lambda = 0, \ldots, 3$ hyperedges $uvwz$. Lemma 2 and $U_P - \mathrm{HGP}(\mathcal{H}_0) = \ell/(2m)$ also hold on three further squares of orders two and three.

c) Source problem of part (a).: A partial Latin square of order n is an $n \times n$ array whose filled cells hold symbols from $\{1, \ldots, n\}$, with no symbol repeated in a row or a column, and deciding whether it can be completed to a Latin square is NP-complete [50]. A triangle ijk of the defect graph has one node in each part and fills the empty cell ij with a symbol

k that row i and column j both lack, so sets of edge-disjoint triangles correspond to conflict-free partial fillings. A row lacks as many symbols as it has empty cells, and so does a column.

d) Size of the witness.: Each side of the switch identity covers 36 pairs and each side of the absorber identity covers 60. For τ triangles, $\mathcal{H}_0$ has $|E_G| + 22\tau$ hyperedges on $|V_G| + 2|E_G| + 28\tau$ nodes, where the count includes nodes of V_G that lie in no hyperedge, and $h_{\min} = \mathrm{HGP}(\mathcal{H}_0) = U_P - 8\ell/(r^2 m)$, which is $U_P - \ell/(2m)$ at $r = 4$.

e) Linearity of the witness.: Every hyperedge of $\mathcal{H}_0$ other than $s\,t\,x_{st}\,x_{ts}$ contains at most one node of V_G or of the nodes x_{st}, and its other members are private to one triangle. Two hyperedges therefore share at most one node unless both are built for the same triangle, and two hyperedges $s\,t\,x_{st}\,x_{ts}$ share at most one endpoint. For one triangle, the six hyperedges on the left of a switch identity pairwise share exactly one node, hyperedges built for different edges share at most a node of V_G, and the seven absorber hyperedges of $\mathcal{H}_0$ pairwise share at most one node. The switch hyperedges contain no node z or f_i, and none of the seven absorber hyperedges of $\mathcal{H}_0$ contains both y_{st} and y_{ts} or a node of V_G together with a node y_{st}, so an absorber hyperedge and a switch hyperedge share at most one node. Hence $\mu_{uv} \le 1$ on $\mathcal{H}_0$.

f) Neighbors in Γ.: A node of V_G is adjacent in Γ to the nodes x of its edges, to the nodes p, y_{st}, y_{ts} of the switches in which it is the smaller endpoint s, to the nodes $q_1, q_2, o_1, o_2, y_{st}, y_{ts}$ of the switches in which it is the larger endpoint t, and to z and two nodes f_i of each of its triangles, where the three nodes of a triangle receive disjoint pairs of nodes f_i. Among the common neighbors of an edge st outside V_G, no hyperedge of $\mathcal{H}_0$ contains a node x together with a node y or z, a node z together with a node y, or nodes private to different triangles, so the adjacent pairs are those stated in the proof.

g) Remarks on part (a).: On these instances, $\mu_{uv} \le 1$ for every pair outside the added nodes forbids repeated hyperedges, so the theorem also holds for r-uniform multihypergraphs. An algorithm that decides the problem without the witness also decides it with the witness, so the hardness persists when the compatible input is withheld and only its existence is promised.

h) Part (b) and the random-walk release.: The reduction of part (b) relies on pair counts leaving the incidence volume open, whereas the entries of P sum to $\mathrm{vol}(\mathcal{H})$ and the weights of the normalized clique sum to $\mathrm{vol}(\mathcal{H})/2$.

i) Example of part (b).: As an instance of part (b) with $R = 3$, let G be a triangle, so $M' = 3$, $S^* = 9$, and $B = 6/19$. The compatible inputs are the triangle as one hyperedge, with $\mathrm{HGP} = 1/3 = M'/S^*$, and its three edges as hyperedges, with $\mathrm{HGP} = 1/4$, so $h_{\max} = 1/3 > B$. For a path with two edges, $M' = 2$, $S^* = 6$, and $B = 4/13$, and its single compatible input has $\mathrm{HGP} = 1/4 < B$.

F. Constant-Gap Hardness

After Theorem 4, the body states that approximating $h_{\max}$ within an additive constant that depends only on the hyperedge size is NP-hard. We state this result with its constant and prove it here.

Theorem 5 (Constant-gap hardness)**.** *For every fixed $r \ge 4$, unless $\mathrm{P} = \mathrm{NP}$, no polynomial-time auditor returns on every input of Theorem 4(a) a bound U with $h_{\max} \le U < h_{\max} + \varepsilon_r$, where $\varepsilon_r = 2/(11975\, r^2)$, and none returns on every such input an estimate U with $|U - h_{\max}| < \varepsilon_r/2$. In particular, $h_{\max}$ has no polynomial-time additive approximation scheme unless $\mathrm{P} = \mathrm{NP}$.*

Proof. For a graph F of maximum degree three with node set V_F, $n_0 = |V_F|$, and edge set E_F, deciding whether its independence number $\mathrm{ind}(F)$ exceeds $\xi_1 n_0$ or lies below $\xi_0 n_0$ is NP-hard for the rationals $\xi_1 = \frac{2M/k+14-2\epsilon}{4M/k+32}$ and $\xi_0 = \frac{2M/k+13+2\epsilon}{4M/k+32}$ of Chlebík and Chlebíková [52, Section 4, p. 19], where $\epsilon > 0$ is arbitrarily small and M and k are parameters of the amplifier that they use, with $M/k \le 40.4$, so that $\xi_1 - \xi_0 = (1 - 4\epsilon)/(4M/k + 32) > 1/200$ for $\epsilon \le 0.007$. Let F' replace every edge of F by a path with six inner nodes. This gives $\mathrm{ind}(F') = \mathrm{ind}(F) + 3|E_F|$, as an independent set I of F extends by the second, fourth, and sixth inner node of each edge $\hat{u}\hat{v}$ counted from $\hat{u}$ if $\hat{u} \in I$ and by the first, third, and fifth otherwise, and conversely an independent set J of F' has at most three inner nodes on each path and at most two if both ends lie in J, so deleting one end of each such path from $J \cap V_F$ leaves an independent set of F with at least $|J| - 3|E_F|$ nodes.

We embed F' in $A_0 \times A_1 \times A_2$ for disjoint sets A_i such that two points agree in exactly two coordinates if the nodes are adjacent and in at most one otherwise. A node $\hat{w}$ of F gets fresh values $\hat{w}^0, \hat{w}^1, \hat{w}^2$, and its at most three edges get distinct ports in $\{0, 1, 2\}$. For an edge with port i at $\hat{u}$ and j at $\hat{v}$, choose permutations (i, i_2, i_3) and (j_1, j_2, j) of $\{0, 1, 2\}$ with $j_1 \ne i$, and walk from the point of $\hat{u}$ by changing coordinates $i, i_2, i_3, i, j_1, j_2, j$ in turn, the first four to fresh values and the last three to $\hat{v}^{j_1}, \hat{v}^{j_2}, \hat{v}^{j}$. The first six points are the inner nodes and the last is $\hat{v}$. Consecutive directions differ and no coordinate returns to an earlier value, so consecutive points agree in exactly two coordinates and points at distance two or more on a walk in at most one. The values of a node occur exactly at that node and on the walks of its edges, and an inner point carries values of at most one node, and two values of it exactly when it is the first or last inner point. Two such points at a node $\hat{w}$ lack the values of $\hat{w}$ in different coordinates and agree in one coordinate, so points on different walks agree in two coordinates only for a node and its adjacent inner point.

Let G be the tripartite graph on $A_0 \cup A_1 \cup A_2$ whose edges are the pairs of values that occur together in a point, and T^* the triangles spanned by the points. We build the instance of part (a) with $V_1 = A_0$, $V_2 = A_1$, and $V_3 = A_2$, and with the hyperedges of the switch and absorber identities only for the triangles of T^*, which gives $m = |E_G| + 22|T^*|$ hyperedges. The argument of part (a) applies to each edge and triangle separately. It shows that $\mu_{uv} \le 1$ on the witness, that the compatible inputs partition the edges of Γ into 4-

cliques, and that the 4-cliques with three nodes of V_G are the hyperedges $uvwz$ with $uvw \in T^*$. The hyperedges $s\,t\,x_{st}\,x_{ts}$ and Y_{st} whose edge st meets V_1 have $\sigma_e = 1/4$ and all other hyperedges have $\sigma_e = 0$, and each hyperedge $uvwz$ covers two such edges. A compatible input with λ hyperedges $uvwz$ has $\mathrm{HGP} = U_P - (\theta - 2\lambda)/(4m)$, where θ is the number of edges of G at V_1. The triangles of these λ hyperedges are edge-disjoint, so $\lambda \le \nu$, and edge-disjoint triangles can be switched and absorbed independently, so $h_{\max} = U_P - (\theta - 2\nu)/(4m)$ for the largest number ν of pairwise edge-disjoint triangles in T^*.

Two triangles of T^* share an edge of G exactly when their nodes are adjacent in F', and no edge lies in three triangles, as F' has no triangle. Hence $\nu = \mathrm{ind}(F) + 3|E_F|$, $|T^*| = n_0 + 6|E_F|$, $|E_G| = 3|T^*| - 7|E_F|$, and $m = 25n_0 + 143|E_F| \le 239.5\,n_0$. With $B_i = U_P - (\theta - 6|E_F| - 2\xi_i n_0)/(4m)$, we have $h_{\max} > B_1$ if $\mathrm{ind}(F) > \xi_1 n_0$, $h_{\max} < B_0$ if $\mathrm{ind}(F) < \xi_0 n_0$, and $B_1 - B_0 = (\xi_1 - \xi_0)n_0/(2m) > 1/95800$. A bound U with $h_{\max} \le U < h_{\max} + 1/95800$ then decides the source problem by testing $U \ge B_1$, and an estimate with $|U - h_{\max}| < 1/191600$ by comparing it with $(B_0 + B_1)/2$. For $r \ge 5$, padding every hyperedge with the same $r-4$ nodes of group b in C_1 as in part (a) multiplies every slack by $16/r^2$ and keeps m, so the gap exceeds $16/(95800\,r^2) = \varepsilon_r$. □

a) Remarks on Theorem 5.: The proof applies to r-uniform multihypergraphs as well, because $\mu_{uv} \le 1$ outside the added nodes forbids repeated hyperedges. The NP-completeness of deciding $h_{\max} = U_P$ rests on part (a) of Theorem 4. The constant results from the 22 hyperedges per triangle of T^*, from $m \le 239.5\,n_0$, and from the gap of the source problem.

We list in Table VII the verdicts at the shared budget B without the two CAHC-init cells, on the ten cells of race and disability in ACS-PUMS-HH, which have less than 30% of their memberships in hyperedges with two or more groups, and for each method. Without the CAHC-init cells, both surrogates miss all six failing cells, and 11 audits are sent to exact evaluation. All ten screened cells pass B, neither surrogate errs on them, and the median ratio of U_P to true HGP is 1.017.

TABLE VII
VERDICTS AT THE SHARED BUDGET B ON FURTHER SETS OF CELLS. THE SCREENED CELLS AUDIT RACE AND DISABILITY IN ACS-PUMS-HH WITH THE FIVE METHODS, THE METHOD ROWS COVER THE 75 CELLS, AND CE+FL IS CE+FAIR-LOUVAIN.

Set of cells	Cells	Fail	U_P certifies passing ↑	$U_{Q,50}$ certifies passing ↑
All 75 cells	75	6	62 of 69	57 of 69
Without CAHC-init	73	6	62 of 67	57 of 67
Screened race and disability	10	0	10 of 10	8 of 10
AHRC	15	1	12 of 14	11 of 14
AHCKA	15	1	13 of 14	13 of 14
GRAC	15	0	15 of 15	14 of 15
CAHC	15	2	11 of 13	9 of 13
CE+FL	15	2	11 of 13	10 of 13